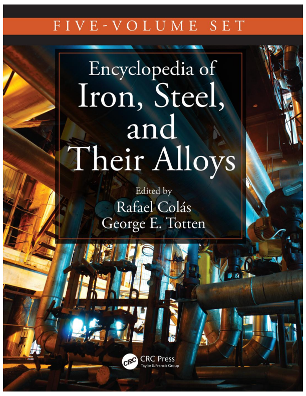

Encyclopedia of Iron, Steel, and Their Alloys

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Advanced High-Strength Steels: Bake Hardening

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Advanced High-Strength Steels: Bake Hardening

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Abstract

Dual phase (DP) steels offer a combination of tensile properties such as low yield strength and a high tensile strength, which make them unique among advanced high strength steels (AHSS). They also exhibit high work hardening rates in the early stage of plastic deformation as well as good bake hardening (BH) potential. The utilization of thermo-mechanical-controlled processing (TMCP) by hot rolling is an industrial method for manufacturing DP steels. In this technique, the control of the rolling parameters and chemical compositions has a significant role on the final microstructure and mechanical behavior.

This entry provides a detailed study of replication of hot rolling mill by TMCP aimed at further development of new categories of DP steels in terms of mechanical properties and bake hardenability via affordable addition of alloying elements together with optimization of the processing conditions. The influence of hot rolling parameters (deformation temperatures and amounts of strain), cooling conditions during $\gamma \rightarrow \alpha$ transformation, amount of martensite as well as variation of alloying elements (C, Si, Mo, and Nb) on the microstructure development, mechanical properties, and BH behavior of DP steels is investigated.

Additionally, this entry offers a detailed study of the local aging and BH behavior of DP steel. For this purpose, two methods were investigated to achieve local strengthening. These methods are local deformation by embossing and local heat treatment by laser.

Furthermore, the influence of load paths and pre-straining on the BH behavior of the DP steel was studied. DP steels were pre-strained under uniaxial, biaxial, and plane strain conditions with different degrees of pre-strain.

INTRODUCTION

To combine fuel saving with increased safety of vehicles, the automotive industry was led to develop AHSS. Their high yield and tensile strengths enable a decrease in sheet thickness (weight saving) and, at the same time, maintain or even improve the crash behavior (safety).[1] This is achieved by micro-alloying[2] and a thermo-mechanical treatment.[3] The group of promising AHSS includes various steel grades such as dual phase (DP), transformation induced plasticity (TRIP), and complex phase (CP) steels. Their microstructures consist of different phases, including ferrite, martensite, bainite, and retained austenite, making up "multiphase" steels.

Multiphase steels are characterized by a good formability, high strength, and a good compromise between strength and ductility. Moreover, these steels exhibit a continuous yielding behavior, low yield point, and a high strain-hardening coefficient.[4,5] This has been attributed to an increase in the work-hardening (WH) limits through forming mobile dislocations due to the martensite transformation during heat treatment and martensite formation in twinning.[6,7]

Furthermore, multiphase steels often show a large potential for BH. BH refers to the increase in yield strength that occurs as a result of the paint baking treatment of the formed auto-body parts. The primary mechanism that causes the additional strengthening is the immobilization of dislocations by the segregation of interstitial atoms, known as classical static strain aging.[8] The increase of strength thus achieved allows a further reduction of sheet thickness and improves the crash safety and the dent resistance. The BH of special steel qualities is technically used in multiphase, where, e.g., the increase in strength is realized in the final heat treatment.[9] Previous investigations[10,11] stated that the BH effect of multiphase is much stronger than that one for conventional BH steels.

Recent investigations showed[12,13] that beside the painting conditions (temperature, time, pre-straining) varying processing parameters during hot strip production has a major influence on the microstructures and mechanical properties of steels and consequently on their bake hardenability. However, the study of the influence of those parameters on the microstructure development, mechanical properties, and BH behavior of AHSS steels has not been sufficiently investigated and is not available in the literature. Thus, the aim of this study is to develop hot-rolled AHSS steels indicating improved mechanical properties and bake hardenability by employing the optimized

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chemical compositions and microstructural constituents together with the feasible processing schedules. For this purpose, the influence of processing parameters, microstructural constituents, and chemical composition on the mechanical properties and the BH behavior in hot-rolled DP steels is investigated. Furthermore, this study is intended to contribute to the understanding of the microstructural evolution during TMCP.

Additionally, this entry presents the study of the local aging and BH effect of hot-rolled DP steels. With a local, restricted initiation of the BH effect, the properties of an originally homogenous material can be locally changed with the aim of adapting them to the required load. As the BH effect significantly depends on the degree of deformation and the downstream thermal treatment,[14,15] it is possible to obtain ductile and high-strength neighboring regions within one component without using sophisticated joining-technologies to combine different materials. Based on these basic investigations, processes are to be developed and optimized to compute this effect as well as to exploit their potential for actual use in high-strength structures.

This entry offers a packet including the results of the activities undertaken regarding the utilization of the BH effect of AHSS in the following chapters:

Chapter 1: Influence of thermo-mechanical process parameters on the mechanical properties and bake-hardening effect of the hot-rolled DP steel

- Influence of hot rolling parameters
- Influence of martensite content and cooling rate
- Influence of the chemical composition

Chapter 2: Influence of the local use of aging and bake-hardening effects in DP steels

Chapter 3: Influence of load paths and bake-hardening conditions on the mechanical properties of the DP steel

CHAPTER 1: INFLUENCE OF THERMO-MECHANICAL PROCESS PARAMETERS ON THE MECHANICAL PROPERTIES AND BAKE-HARDENING EFFECT OF THE HOT-ROLLED DP STEEL

For hot strip rolling of steels it is the desired final mechanical and geometrical properties of the strip that determines the rolling schedule, which is the setup for the rolling process, i.e., the amount of reduction, the temperature, cooling conditions, chemical composition of the material in the current work, a wide range of influencing parameters is used to clarify the effect of different TMCP schedules on the microstructure development, mechanical properties and BH behavior of the DP steel. The optimized TMCP schedules are discussed in relation to the microstructure evolution and mechanical properties as well as the BH behavior.

Table 1 Chemical composition of the steel (wt.%).

Steel	DIN 10336	C	Si	Mn	Cr	P	N	Al
DP 600	HDT580X	0.06	0.10	1.30	0.60	0.04	0.006	0.03

Materials and Experimental Methods

The DP steels used for this entry was produced industrially by Salzgitter Flachstahl GmbH as roughing rolled plates (including melting, casting, and roughing rolling) with a thickness of 50 mm. The chemical composition of the steel is listed in Table 1.

Simulation of Finishing Rolling

For studying the kinetics of phase transformations taking place after TMCP and the last three hot rolling deformation stages, a “Bähr TTS820” type deformation simulator was used. The experiments were performed using a flat compression setup of the deformation simulator (Fig. 1). The dimensions of the flat compression sample are shown in Fig. 2. Heat transfer to the shoulders of the flat compression samples was reduced by two holes.

In the experimental setup, the flat compression specimen is placed on ceramic rollers and fixed from the upper side by two ceramic rods. The specimen is inductively heated by a U-shaped induction coil. Two deformation punches are used to deform the specimen in plane strain compression. During compression, the two ceramic rods move upward allowing the sample to extend in its longitudinal direction. Four gas nozzles with holes facing to the center of the specimen are used to quench the sample (Fig. 1). Helium gas is used for cooling. The equipment has a laser extensometer, which measures the dilation of the sample width during the experiments. Sheathed type S Pt/Pt-10% Rh thermocouple wires with a nominal diameter of 0.2 mm are individually spot welded in the center of the sample’s surface.

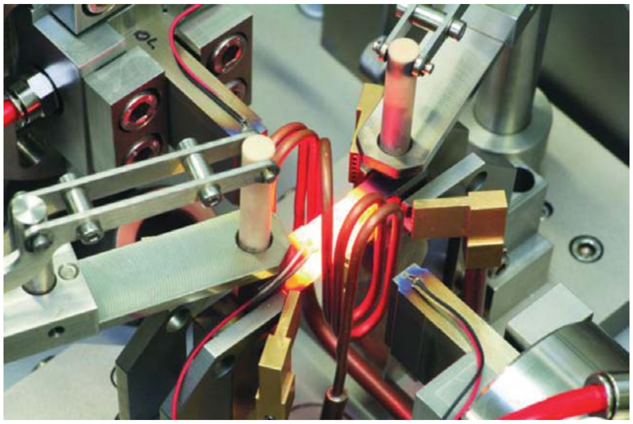

Fig. 1 Experimental flat compression setup and position of the flat compression sample in the deformation simulator.

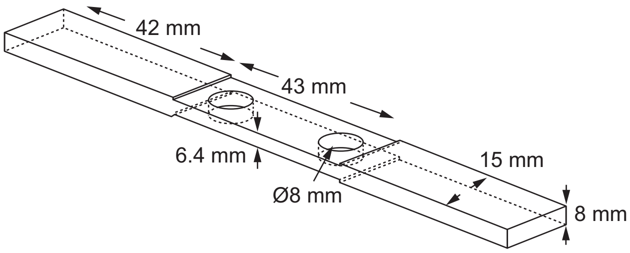

Fig. 2 Flat compression test; sample dimensions.

Tensile Testing and Bake-Hardening Treatment

Tensile specimens with special geometry were machined out of flat compression specimens to determine the mechanical properties. Fig. 3 shows flat compression specimens before and after hot deformation as well as after machining for tensile testing. Minimum three specimens were tested for each condition and the results were averaged. The divergence of temperature in the deformed zone recorded was ±7 K. A quite homogenous microstructure could be observed in the deformation zone. The small tested zone of the sample (($t \times 8 \times 10$) mm) is appropriate for the observed fine microstructure (grain size below 10 μm) to give satisfactory reproducibility of the results. This is revealed in the reasonable standard deviation of the observed mechanical properties as will be seen.

Tensile tests were conducted using a universal testing machine (UTS) with a 250kN load cell and a crosshead speed of 5 mm/min. An extensometer with a gauge length of 15 mm was used.

The BH was determined using the procedure described in the standard SEW 094.[16] The samples were pre-strained with $\varepsilon = 2\%$ in tension. After pre-straining (PS) they were heat treated in an open air electric furnace at a temperature of 170°C for 20 min and cooled in air. After the BH heat treatment, the samples were strained in tension until fracture.

BH_2-values were calculated as the difference between the yield stress after the paint baking and flow stress at the end of the PS. For the samples tested without PS the difference between yield strength ($R_{p0.2}$) and lower yield strength (R_e) of the respective BH sample was taken as a measure of BH_0. A minimum of three samples were tested for each condition and the averaged results are reported in the present work.

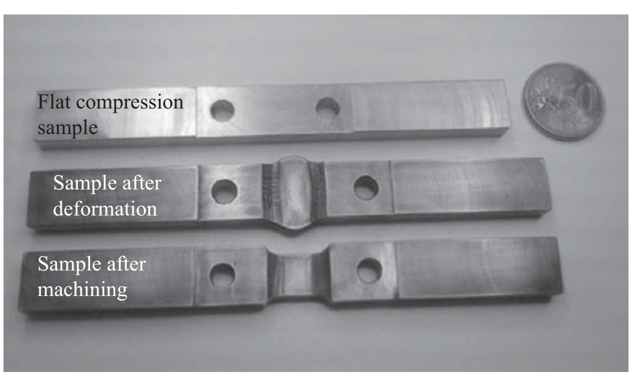

Fig. 3 Flat compression specimens before and after hot rolling simulation and after machining.

Microstructure Investigations

The microstructures of thermo-mechanically produced samples were investigated using light optical microscopy (LOM) and transmission electron microscopy (TEM). In addition, high-angle annular dark field (HAADF) imaging was used in scanning transmission electron microscopy (STEM). For the LOM preparations, the samples were etched with Nital. TEM was carried out using a JEM-200CX TEM operating at 200 kV. TEM and STEM specimens were prepared from 3 mm disks which were ground and polished to a thickness less than 0.1 mm to reduce the magnetic mass of the specimens.[17] The disks were electro-polished using a standard double-jet procedure (TunelPol-5, Struers). An electrolyte consisting of 5% perchloric acid ($HClO_4$) in methanol (CH_3OH) was used. Electro-polishing was done at −30°C using an operating voltage of 50 V.

Influence of Hot-Rolling Parameters

Aim of the study

Prior to finishing, the austenite grain size will vary, depending on the amount of reduction and the finishing temperature. Hence, in the current work, a wide range of finishing strains and temperatures are used to clarify the effect of different TMCP schedules on the phase transformation kinetics, microstructure development, mechanical properties, and BH behavior of the DP steel. The optimized TMCP schedules are discussed in relation to the microstructure evolution and mechanical properties as well as the BH behavior.

Thermo-mechanical-controlled process schedule

In order to influence the shape and size of the austenite grains before $\gamma \rightarrow \alpha$ transformation, austenite conditionings were conducted using three different deformation schedules. With the estimated non-recrystallization temperature T_{nRX}, the deformation part for the schedules was determined in such a way that all the three possibilities were covered, namely all deformations conducted above T_{nRX}, deformations below T_{nRX}, and deformations mixtures of above and then below T_{nRX} (named above–below T_{nRX}).

Fig. 4 illustrates schematically different schedules applied in this entry. The data corresponding to the numbers denoted on this figure are collected in Table 2. The upper and lower limits of technological influencing parameters were selected according to industrial processes. The

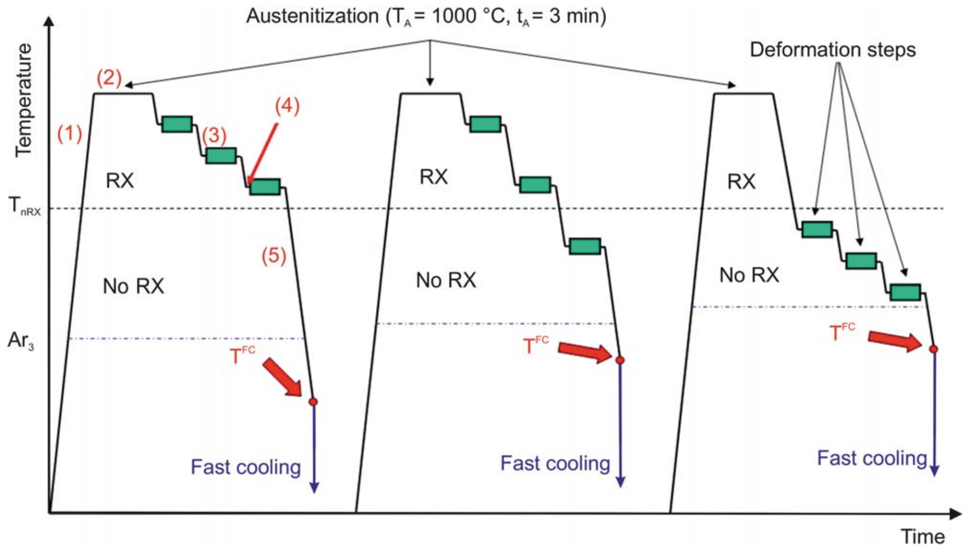

Fig. 4 The schedules used for simulation of the final steps of finishing hot-rolling process, recrystallization region (RX), and non-recrystallization region (No RX). (1)-(5) see Table 2.

finishing temperatures (T_f) in large-scale production of hot-rolled DP steels are between 780°C and 880°C. Therefore, the deformation temperatures and the amounts of strain varied close to this interval.

After austenitizing at 1000°C for 3 min, flat compression specimens were subjected to three defined deformations in three different temperature intervals. Strain rate of each deformation step was $\dot{\varphi} = 10\,\text{sec}^{-1}$. Two cooling stages were performed during TMCP. After the last deformation step, the specimens were cooled to the fast cooling start temperature (T^{FC}) with 10 K/sec until required fraction of ferrite was obtained ($\gamma \rightarrow \alpha$ transformation). Then, specimens were accelerated cooled to room temperature with a high cooling rate of ~100 K/s to achieve martensite from retained austenite ($\gamma \rightarrow \alpha'$ transformation). The amount of martensite in industrially produced DP steels is typically between 10% and 30%. However, the total martensite content can extend as high as 50–60–80% also in DP 800/DP1000/DP1200 steels. Many studies have shown that the martensite volume fraction (MVF) is dominant in controlling the tensile properties. The MVF is basically varied by changing the intercritical annealing temperature. Experimental results from literature state that increasing the martensite volume fraction increases the ultimate tensile strength while the elongation decreases.[18,19] The effect of varying MVF is introduced in section "Influence of Martensite Content and Cooling Rate."

Table 2 Data corresponding to Fig. 4.

Denotation	Specification		Corresponding data
(1)	Heating rate	(K/sec)	10
(2)	Austenitization time	(s)	120
(3)	Cooling rate before each deformation	(K/sec)	10
(4)	Break time before each deformation	(sec)	5
(5)	Cooling rate	(K/sec)	10
	Strain rate of each deformation step	(sec^{-1})	10
T^{FC}	Start of fast cooling	(°C)	To be defined

In this section, a fixed martensite volume fraction MVF = 20% was chosen. Minimum three DP samples with prescribed amount of ferrite (80%) and martensite (20%) were prepared for each schedule. Table 3 illustrates the selected schedules for TMCP simulation. The control of the phase distribution and determining T^{FC} are described in the following section as a part of results.

Results

Phase transformation behavior and determining $\mathbf{T^{FC}}$**.** In order to obtain DP steels with defined amount of phases the $\gamma \rightarrow \alpha$ transformation kinetic must be known. Then the appropriate T^{FC} can be found from which the specimens have to quench below M_S. In order to determine T^{FC} temperatures depending on applied schedules, deformation/dilatometric tests had been performed using the deformation simulator. The specimens were prior subjected to the same deformation schedules as TMCP (Table 3) and subsequently cooled from the last deformation step to RT at a cooling rate of 10 K/sec, which is the same as the first cooling stage of TMCP. The results of dilatometric measurements are shown in Fig. 5A where change in length is plotted vs. temperature during cooling. For better discretization, the dilatation curves of three schedules (1, 6, and 11) are plotted in this figure to show the shifting of the $\gamma \rightarrow \alpha$ transformation nose toward shorter incubation

Table 3 Hot deformation schedules and values of influencing parameters for each schedule; φ_t is the amount of total strain.

Number of schedule	T_1 (°C)	T_2 (°C)	T_3 (°C)	φ_1	φ_2	φ_3	φ_t
Sch. 1				0.45	0.30	0.20	0.95
Sch. 2				0.45	0.25	0.15	0.85
Sch. 3	930	900	855	0.40	0.25	0.15	0.80
Sch. 4				0.35	0.20	0.20	0.75
Sch. 5				0.35	0.20	0.10	0.65
Sch. 6				0.45	0.30	0.20	0.95
Sch. 7				0.45	0.25	0.15	0.85
Sch. 8	900	855	830	0.40	0.25	0.15	0.80
Sch. 9				0.35	0.20	0.20	0.75
Sch. 10				0.35	0.20	0.10	0.65
Sch. 11				0.45	0.30	0.20	0.95
Sch. 12				0.45	0.25	0.15	0.85
Sch. 13	855	830	800	0.40	0.25	0.15	0.80
Sch. 14				0.35	0.20	0.20	0.75
Sch. 15				0.35	0.20	0.10	0.65

times and higher transformation temperatures for deformations at lower finishing temperatures.

The transformed austenite fraction was calculated by the lever rule from the variation of the change in length as a function of temperature. The progresses of $\gamma \rightarrow \alpha$ phase transformation had been determined by measuring length change of samples as a function of temperature. From those measurements the austenite fraction f_γ was calculated according to the following equation:

$$f_\gamma = \frac{y}{x + y} \tag{1}$$

where x and y are length change functions of temperature for the nonisothermal cooling experiments as shown in Fig. 5.

The calculation results are given in Fig. 5B showing the austenite volume fraction as a function of temperature. From this figure, T^{FC} temperatures for different schedules can be found. The data of Fig. 5 indicate that the hot deformation temperatures exert a stronger influence on the Ar_3 and Ar_1. The highest Ar_3 and Ar_1 temperatures are achieved for schedules applied below T_{nRX}. It can also be seen that increasing the total amount of strain for each temperature interval results in increasing $\gamma \rightarrow \alpha$ transformation temperatures (Ar_3 and Ar_1) and T^{FC}.

Microstructure evolution. Microstructural evolution of thermo-mechanically produced DP specimens with defined f_α and f_α was studied. Fig. 6 displays exemplary the microstructure of two DP steels subjected to schedule 1 (deformed above T_{nRX}) and schedule 11 (deformed below T_{nRX}). The Nital etchant reveals the martensite dark, while the ferrite remains white. During the first cooling stage after the last deformation step austenite progressively transforms to ferrite, whereas the remaining part transforms to martensite. All images show a classical DP microstructure with relatively globular martensite islands embedded in the ferrite matrix phase. The ferrite

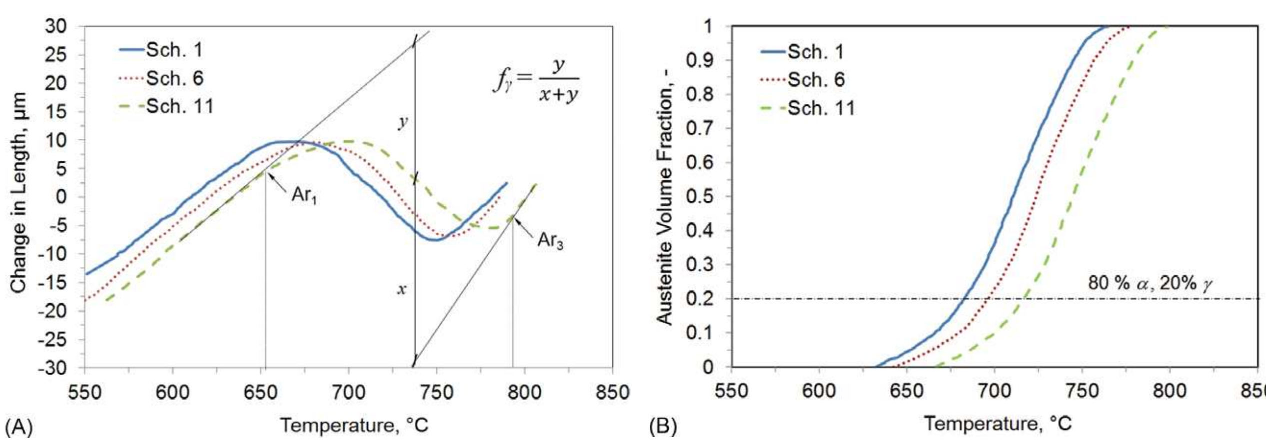

Fig. 5 Influence of the hot deformation schedule on the phase transformation behavior; (A) dependences of the change in length on the temperature at $\gamma \rightarrow \alpha$ phase transformation during cooling stage at 10 K/sec and (B) calculated fraction of $\gamma \rightarrow \alpha$ as a function of temperature for specimens applied different hot deformation schedules.

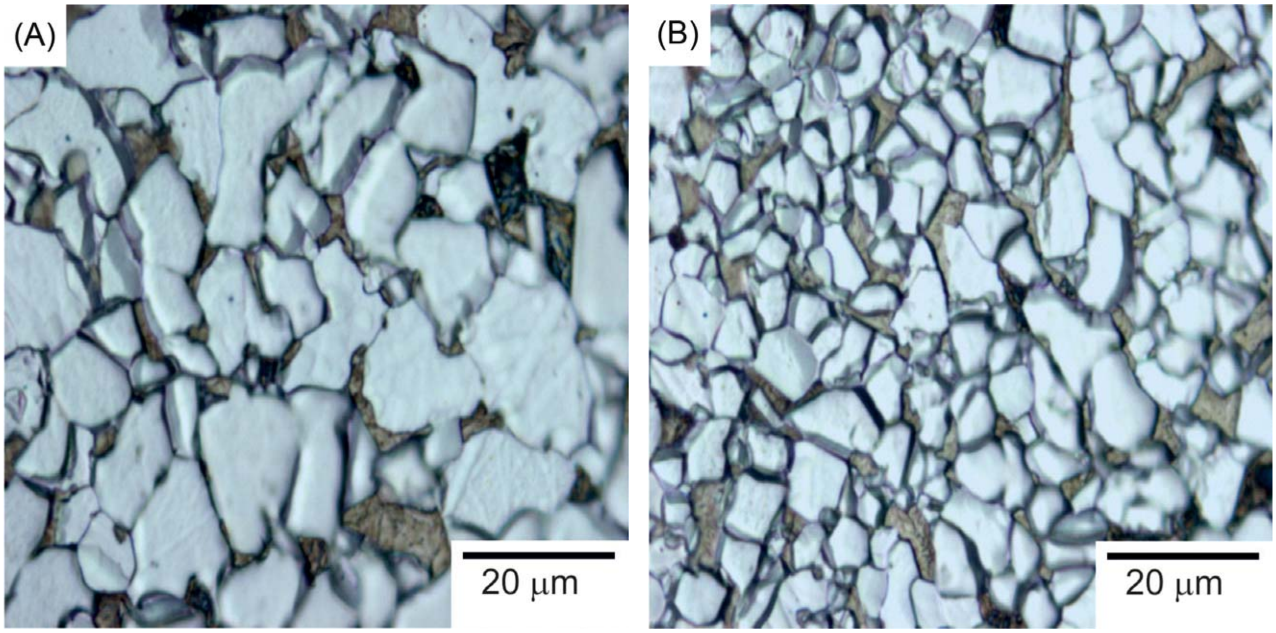

Fig. 6 Microstructure of DP steels showing different ferrite grain sizes and martensite blocks obtained after TMCP when all the deformation steps were conducted: (A) above T_{nRX} (Sch. 1) and (B) below T_{nRX} (Sch. 11); cooling rate of all samples in first cooling stage = 10 K/sec and in second cooling stage = 100 K/sec.

grains are equiaxed with average sizes depending on the applied hot deformation schedule. The MVF determined by the line intercept method is ~20% for all samples. Small amounts of retained austenite between 1% and 2% were found by saturation magnetization measurements for all conditions. Martensite islands can be clearly observed in the microstructure. They often display dark substructures either within or in their immediate surroundings. In addition, such a dark phase can also be observed at the boundaries between two neighboring ferrite grains.

The influence of hot deformation schedule on the microstructure is very significant. It can be seen from Fig. 6 that deformation of the steel in the non-recrystallization region results in a finer grained structure, which is the expected result.

By deforming the material in the recrystallization region the austenite grains are being refined. These are important because the grain size of the austenite strongly affects both the kinetic of subsequent transformation from austenite into ferrite and the ferrite grain size, namely smaller austenite grains consequently lead to the refinement of ferrite grains. When deformations are applied at temperatures below T_{nRX}, the austenite grains become elongated and deformation bands are introduced within the grains. The process is called pancaking. As the amount of deformation in this region increases, the number of nucleation sites at the austenite grain boundaries and within austenite grains increase. Because of that, the austenite–ferrite transformation from deformed austenite is accelerated (Fig. 5) and yields much finer ferrite grains than that from recrystallized, strain-free austenite (Fig. 6).

Further investigations in the microstructure show that varying TMCP parameters influence not only the grain size but also the morphology of ferrite and martensite. Fig. 7 compares the development of dislocation distributions of two DP steels following schedules 1 (deformed above T_{nRX}) and 11 (deformed below T_{nRX}) analyzed by HAADF STEM. It can clearly be seen that the dislocation density significantly increases with deformation in the non-recrystallization region. An evidence of low dislocation density is observed for schedule. 1 when the sample is deformed above T_{nRX} at the highest amount of total strain (Fig. 7A), while for schedule 11 the dislocation density is seen to be increased (Fig. 7B). Additionally, a strongly heterogeneous structure with tangles can be seen in the ferrite matrix, though the density of the tangles increases in and around the grain boundary region. For both conditions dislocations inside the ferrite grains are distributed irregularly. While in the interior of the grains usually a relatively low dislocation density is observed, at the F–M interfaces a strongly increased number of dislocations is present. This is due to the volumetric expansion from austenite to martensite by accelerated cooling during TMCP

Mechanical properties. The results of the basic tensile tests of all DP steels with 80% ferrite and 20% martensite, produced by TMCP, are given in Fig. 8. This figure demonstrates the effect of hot deformation parameters on the R_m, $R_{p0.2}$, and TEl with their accompanying standard deviations for all specimens. Evaluating the different conditions of DP steels, a pronounced influence of the

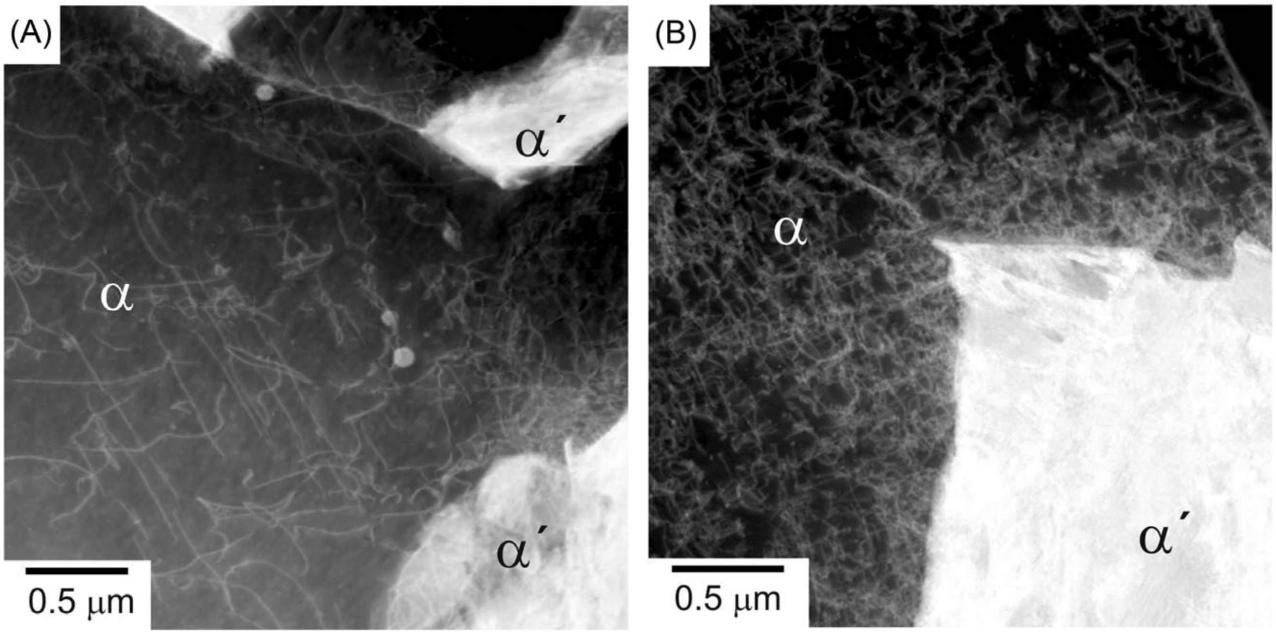

Fig. 7 HAADF STEM images of DP steels showing a good view of dislocation density within the ferrite grains obtained from: (A) Sch. 1 applied above T_{nRX} and (B) Sch. 11 applied below T_{nRX}. All schedules deformed at the highest amount of total strain (φ_t = 0.95).

hot deformation schedule is obvious. The largest strength (676 ± 9) MPa, but lowest ductility (18.5 ± 3)% can be observed for schedule 11 with the samples been deformed below T_{nRX} at the highest amount of total strain (φ_t = 0.95). For schedule 6, in which the samples are deformed above–below T_{nRX} at the same total strain as before, still quite high strength (656 ± 14) MPa is achieved. At the same time, TEl increases to more than 20%. Compared to schedule 11 a slightly lower $R_{p0.2}$ is observed (432 ± 11 vs. 448 ± 9) MPa. The mechanical properties of schedule 1 show lower values of R_m and $R_{p0.2}$ under this condition. R_m with (642 ± 12) MPa is found to be slightly reduced when the amount of total strain is decreased (schedule 2). Comparing schedule 1 with schedule 11, it can be seen that a larger strength level combined with a lower ductility is found for the last one deformed in the non-recrystallized austenite region. Schedule 5 subjected to the lowest amount of total strain indicates the lowest values of R_m and $R_{p0.2}$. Comparing schedules 11–15 followed hot deformations below T_{nRX} the lowest value of R_m (637 ± 8) MPa and $R_{p0.2}$ (415 ± 3) MPa but the largest value of TEl (22.3 ± 4)% is found for schedule 15 having the lowest amount of total strain. A slightly higher level of R_m, $R_{p0.2}$ is observed for schedules 13 and 14 at higher total strain.

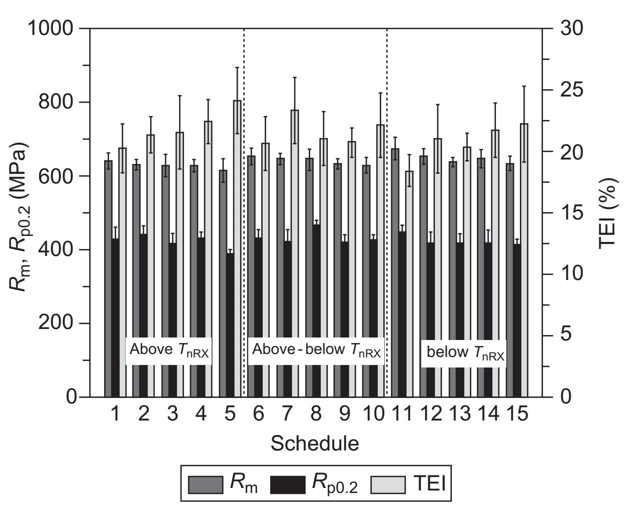

Fig. 8 Tensile strength (R_m), yield strength ($R_{p0.2}$), and total elongation (TEl) dependence on the hot deformation schedules after TMCP.

Bake hardening behavior. Fig. 9 displays BH_0 and BH_2 of DP samples for different TMCP schedules. This is best revealed in this figure showing the comparative results of BH_x (BH increment after PS at 0% or 2% and BH at 170°C for 20 min) increments. Referring to the graphs in Fig. 9, it can be concluded that pre-straining the samples to 2% and BH heat treatment result in relatively high BH values. Furthermore, on average the higher increase of the BH is obtained for samples subjected to deformations in non-recrystallized austenite region at larger amounts of total strain. Schedules followed deformations above–below T_{nRx} exhibit intermediate values.

Discussion

The average values of BH_0 and BH_2 have been calculated from the values recorded at each combination of hot

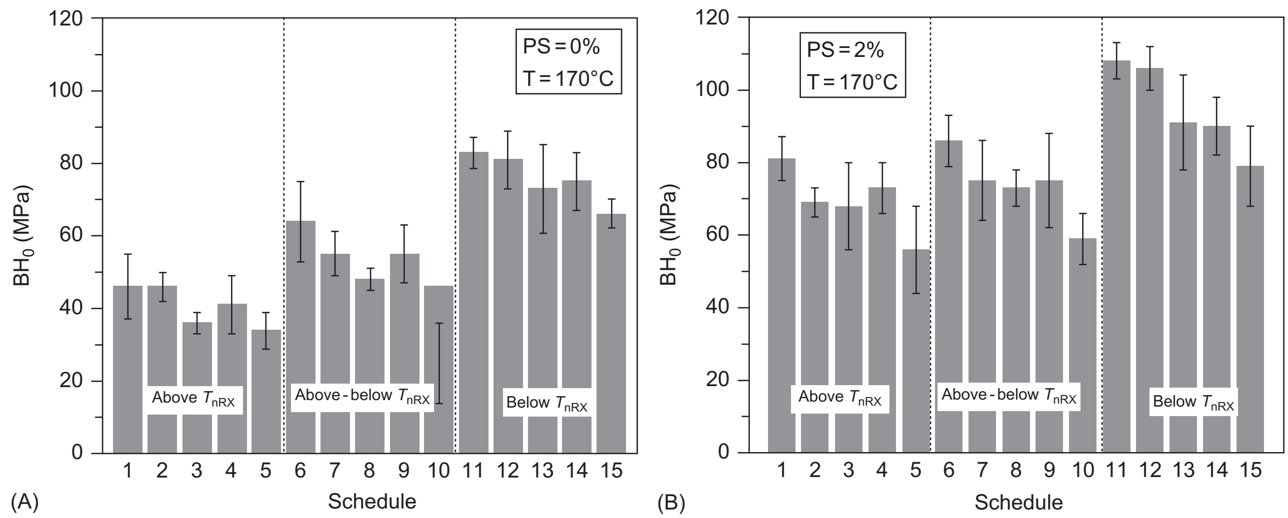

Fig. 9 (A) Dependence of BH_0 on the hot deformation schedules, calculated from difference between $R_{p0.2}$ of the tensile sample and R_e of the respective BH sample and (B) dependence of BH_2 on the hot deformation schedules.

deformation condition. A comparison among these averages is shown in the histogram of Fig. 9. From this figure it is clear that a high BH level is obtained when steels are deformed at high amounts of total strains ($\varphi_t > 0.8$) or at the highest amount of the strain of the last deformation step ($\varphi_3 = 0.2$) or below T_{nRX}. The highest BH_0 and BH_2 values are recorded for schedules following hot deformations in the non-recrystallization region to the highest amount of total strain. The hot deformation schedule has a pronounced effect on the grain size.[13] A deformation below T_{nRX} or at high amounts of total strain results in a refinement of the final DP steel microstructure. Thus, the improvement in BH behavior for these schedules seems entirely to be due to the microstructural refinement.[20] In addition to the effect of grain refinement on the improvement of the bake hardenability, the higher dislocation density of the finer grains has a further improving effect.[18] A lower BH level for the schedules deformed above T_{nRX} to low amounts of total strain could be related to a lower dislocation density and coarse grains after TMCP. For schedules with lower deformation temperatures (below or above–below T_{nRX}) and smaller amounts of total strain, an increasing trend in BH effect with decreasing mean grain size can be identified. This can be probably attributed to the influence of grain size on the concentration profiles of dislocation in the interior of the grains and in the grain boundaries. It is expected that there will be a relatively high dislocation level present in the matrix for a smaller grain size, which is also confirmed by the HAADF STEM images of Fig. 7. This results consequently in more intensive dislocation pinning by free C atoms. Therefore, the BH effect would increase with decreasing grain size because the distance, which the free C atoms move to the grain boundaries, is less.[21,22] However, there are evidences in literatures that grain size becomes important only after a critical size. For instance, Zhao et al.[23] concluded that the influence of segregation of carbon to grain boundaries on the formation of Cottrell atmosphere for grain sizes below 16 μm is increasing with decreasing grain size and is negligible for grain sizes larger than 16 μm. Although this study is carried out for an ultra-low carbon steel, yet the conclusion could be applicable here as well. The grain sizes observed during the current study are below 10 μm, implying the significance of varying grain size on the bake hardenability as expected by the aforementioned hypothesis.

Furthermore, small grain sizes (deformed below T_{nRX}) contain increased dislocation densities after deformation due to the high number of obstacles in form of grain boundaries.[22] Probably, a large part of the solute carbon is present near the F–M interfaces or in the grain boundaries of the ferrite, which may contribute to the process of BH. On the other hand, the diffusion of C atoms toward the energetically favored positions on the dislocation lines is a rather fast process under the given conditions. In the case of a fine grain size, solute C atoms from the grain boundaries can move faster to dislocations in the middle of a grain due to the shorter distances.[22]

From Fig. 9, it can also be noted that BH values after PS = 2% are significantly higher compared to the conditions without PS, where the binding between dislocations and the C atoms is strong. Thus, it is more difficult for the C atoms to diffuse into the ferrite for dislocation pinning. For the C atoms present in the ferrite in a low concentration, it takes a long diffusion path to reach the few dislocations being present. At the same time there are ample dislocations at the F–M interface to maintain smooth yielding without distinct increase of yield strength, although part of them may be anchored by carbon atoms. As soon as the material is pre-strained, the dislocations start to move, releasing C atoms, which now reach the increased number of dislocation lines inside the ferrite more easily resulting in a higher increase of yield strength. When the pre-strained samples were baked, it is generally noticed that the microstructural developments from the tangled array of dislocations to cells occur. Dislocations start to cluster inside individual grains, and these clusters eventually join together to form a cell structure after the baking treatment.

Conclusions

1. By making use of the beneficial effect of the hot deformation below T_{nRX} and/or to high amounts of total strain, pronounced finer ferrite grains and finer martensite blocks in DP steels had been produced. This results in improvement of mechanical properties and BH behavior of the steels.
2. For the given alloying composition, the most promising microstructures with respect to the mechanical properties are those processed below T_{nRX} at higher amounts of total strain. The microstructures of this steel seem to be strongly dislocated, resulting in a heterogeneous structure with dislocations irregularly distributed in the ferrite matrix. A higher dislocation density is observed for the samples applied at higher amount of total strain. This results in improved strength after TMCP as well as after baking process.
3. A noticeable higher strength increment is observed after pre-straining at 2% and baking at 170°C for 20 min as compared to that one without PS. This increment is significantly higher for conditions conducted below T_{nRX} at high amounts of total strain.
4. By controlling the deformation temperatures and the amounts of strain during the TMCP, it is possible to achieve the optimized microstructure and hence to improve the mechanical properties and bake hardenability of the DP steel.

Influence of Martensite Content and Cooling Rate

Aim of the study

This investigation aimed at designing microstructural variations in DP steels and assessing the cooling rate influenced by such microstructural variations. The elements of microstructural control through TMCP in DP steels along with a consideration of microstructural effects on mechanical properties, with emphasis on cooling rate during $\gamma \rightarrow \alpha$ transformation, will be discussed in detail. The following variations in the microstructure were sought in the DP materials for the present study: 1) vary the volume fraction of martensite and 2) vary the cooling rate during $\gamma \rightarrow \alpha$ transformation.

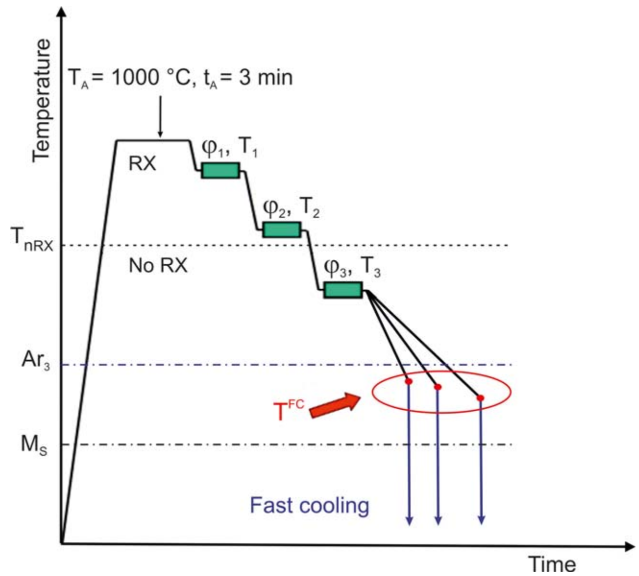

Fig. 10 TMCP schedule applied for simulation of the final steps of finishing rolling process.

Thermo-mechanical-controlled process schedule

Fig. 10 illustrates schematically the employed TMCP schedule for simulating the final steps of the finishing rolling. All flat compression specimens had been austenitized at a temperature of T_A = 1000°C for t_A = 3 min. The heating rate was 10 K/sec. The hot deformation parameters of the last three deformation steps had been selected according to industrial conditions and kept fixed (Table 4). The first and second deformation steps took place in the recrystallization region (above T_{nRX}) followed by a further step in the non-recrystallization region. The strain rate of each deformation step was $\dot{\varphi} = 10\,\mathrm{sec}^{-1}$. The intermediate time between the deformation steps was 5 sec. After the last deformation step the specimens were cooled below M_S in two stages. First, they were cooled at different cooling rates to fast cooling start temperature (T^{FC}) in $\gamma \rightarrow \alpha$ transformation region until required fraction of ferrite and austenite was obtained and then accelerated cooled below M_S at a cooling rate of ~100 K/sec.

In order to obtain DP steels with prescribed amounts of ferrite and martensite, the appropriate T^{FC} temperatures must be determined from which the specimens have to be fast cooled. As discussed in the previous chapter, varying TMCP parameters affects this temperature. Determining

Table 4 Parameters for the simulated hot deformation process according to industrial conditions.

Steel	T_1 (°C)	T_2 (°C)	T_3 (°C)	φ_1	φ_2	φ_3	φ_t
DP 600	900	855	830	0.45	0.25	0.15	0.85

T^{FC} is described in following section. The accelerated cooling of the specimen from T^{FC} below M_S results in transformation of prescribed remaining austenite to martensite. Therefore, adjustment of MVF is possible by variation of T^{FC}. Variation of the cooling rate during the $\gamma \rightarrow \alpha$ transformation and volume fraction of martensite in the DP 600 steel are shown in Table 5.

To define T^{FC} depending on the cooling rate, dilatometric measurements had been performed, as described before (Phase Transformation Behavior and Determining T^{FC}). The detailed description of the phase transformation study including T^{FC} temperatures is introduced in.[24]

Results

Microstructure evolution. Flat compression samples were cooled after the last deformation step to different T^{FC} temperatures at different cooling rates and then accelerated cooled below M_S. The TMCP was designed to produce DP microstructures with different MVFs. LOM investigations were performed to study the microstructure evolution after TMCP. Fig. 11 displays exemplary LOM images corresponding to samples containing 10%, 20%, and 30% of MVF with a cooling rate of 10 K/sec. It is revealed that three DP microstructures are characterized by similar cooling rate (10 K/sec) but exhibit different MVFs. The estimated MVFs in the final microstructure based on dilatation curves are in good agreement with the quantitative analysis. All images show a classical DP microstructure with relatively globular martensite islands embedded in the ferrite matrix phase. Large martensite islands can clearly be observed, but they often show dark substructures either within or in their direct surroundings. In addition, such a dark phase can also be observed at the boundaries between two neighbors. Some bainitic phases could also be detected in the microstructure. Small amounts of retained austenite between 1% and 2% could be analyzed by applying saturation magnetization measurements for all conditions. These observations were studied in the previous work of the authors.[20]

For samples cooled during the $\gamma \rightarrow \alpha$ transformation with different cooling rates and the same MVFs, no significant differences with regard to the morphology, phase distribution, and grain size could be found.

Mechanical properties. The engineering stress–strain curves of DP steels with MVFs of 10%, 20%, and 30% and a cooling rate of 10 K/sec are plotted in Fig. 12.

Table 5 Variation of cooling rate during simulation of controlled cooling after the last deformation to T^{FC} and variation of MVF.

Steel	Cooling rate after last deformation to T^{FC} (K/sec)	MVF (%)
DP 600	1, 10, 20, 30, 40	10, 20, 30

All three steels show a continuous yielding behavior. Rigsbee et al.[25] have stated that at least 4% of martensite in DP steels is necessary to achieve a continuous yielding behavior. Ultimate tensile strength, yield strength, and total elongation of the specimens with different MVFs after TMCP at different cooling rates are compared to the ones after PS and BH treatment as illustrated in the next section. As expected, Fig. 12 shows that both the yield and tensile stress increase with the increasing volume fraction of martensite. At the same time, the elongation decreases. The difference in elongation between the 10% and 30% MVF samples is not significant, which can well be considered within experimental errors. This behavior is commonly interpreted in terms of local dislocation accumulation[26] introduced by the martensitic transformation. Yet, as the plastic strain of the martensite phase is negligible, the total elongation to fracture is reduced with increasing martensite fraction.

Bake hardening behavior. Fig. 13 compares the effects of cooling rate and MVF on the strength increase after PS and BH in terms of BH_0, BH_2, values. At each cooling rate, this figure plots the difference between the yield strength after PS and BH (R_e) and the yield strength of the samples after TMCP ($R_{p0.2}$). As shown in this figure, both BH_0 and BH_2 values increase with increasing MVF at all cooling rates. Without PS the samples indicate a lower BH level than with PS = 2% (BH_2). DP steels with 10% MVF reveal no significant increase in BH_0 values at all cooling rates. For DP steels with 30% MVF, BH_0 increases steeply with increasing cooling rate. For steels having 20% MVF, BH_0 increment is found after a cooling rate of 20 K/sec (Fig. 13A).

The lowest BH_2 value of about 73 MPa is reached for the steel with 10% MVF at the lowest cooling rate of 1 K/sec, while the highest BH_2 value of about 120 MPa could be obtained with 30% MVF and a cooling rate of 40 K/sec (Fig. 13B). DP steels containing 10% MVF appear no distinct BH_2 increment at low cooling rates (less than 10 K/sec), while BH_2 values at high cooling rates show an increased level.

Three major observations from these data on the response to BH are as follows:

1. Pre-strained DP steels to 2% indicate a remarkably higher BH level compared to non-pre-strained steels.
2. At high cooling rates (above 20 K/sec), there is a significant increase in strength due to aging and a return of a well-defined yield point on loading, i.e., the cooling rate sensitivity of the baked materials at the low cooling rates is much lower than the high cooling rates.
3. The contribution of BH increases with an increase in MVF at all cooling rates.

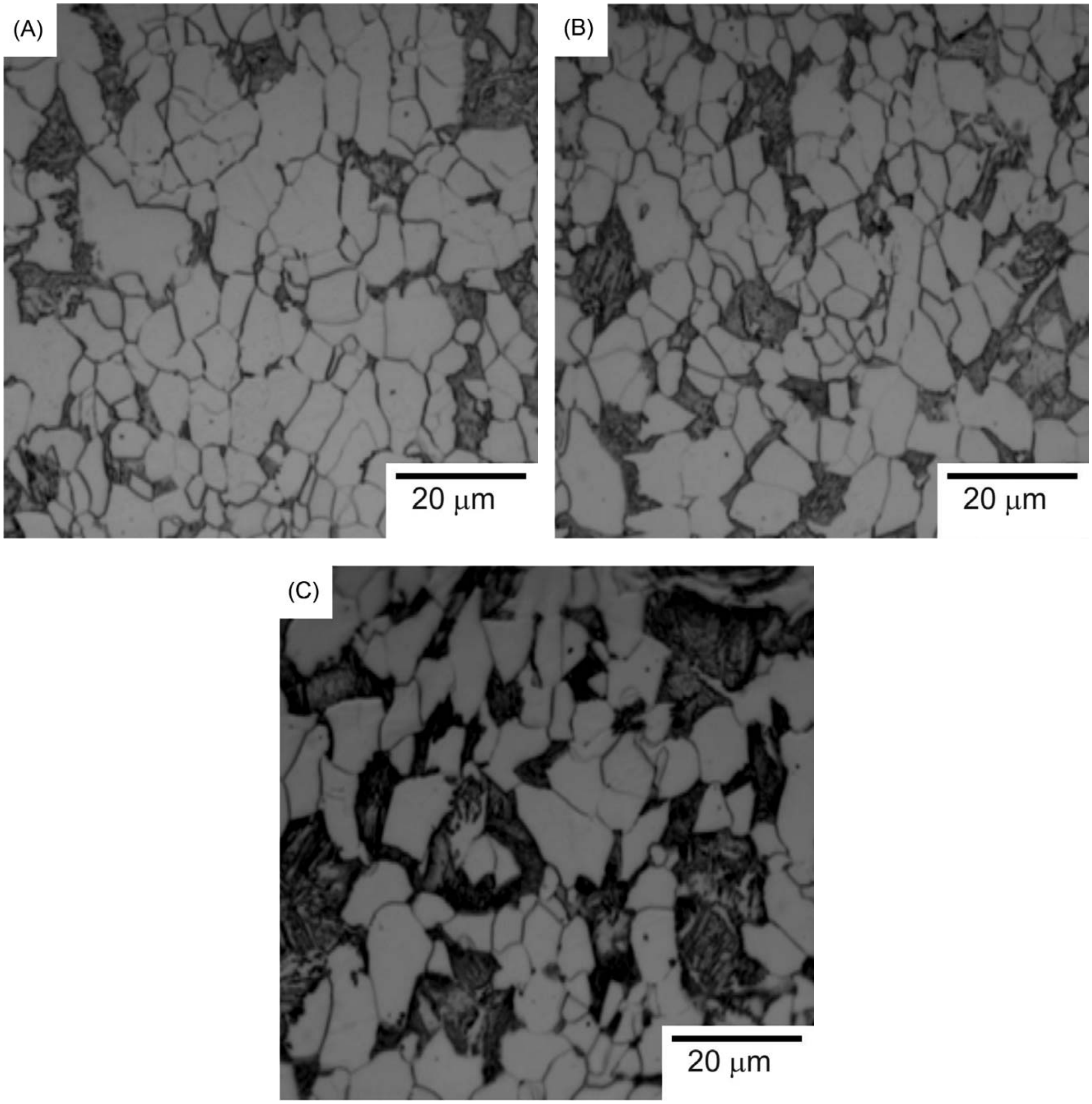

Fig. 11 Microstructure of DP steels showing different volume fractions of martensite obtained after TMCP: (A) $f_\alpha = 90\%, f_{\alpha'} = 10\%$; (B) $f_\alpha = 80\%, f_{\alpha'} = 20\%$, and (C) $f_\alpha = 70\%, f_{\alpha'} = 30\%$; cooling rate of samples in the first cooling stage was 10 K/sec.

Discussion

The increase in the dependency of bake hardening with cooling rate indicates that there is an increase in the contribution of thermally activated dislocation processes, e.g., the formation of double kink pairs, as dislocations move past Peierls barriers, to the overall stress required for dislocation movement such that at high cooling rates the properties are dominated by the stress for individual dislocation movement.[27] The relaxation mechanism can be prevented during thermo-mechanical processing at higher cooling rates after the last deformation. This causes the generation of more dislocation in the final microstructure, which could increase the strength effect in DP steels.[3]

Clearly high BH_0 and BH_2 values are observed as the MVF increases (Fig. 13). Obviously, a large volume fraction of martensite favors large BH without and with 2% pre-strains. However, probably it is not only the MVF which governs this process but rather the state of martensite is likely to influence the BH behavior as well. It is

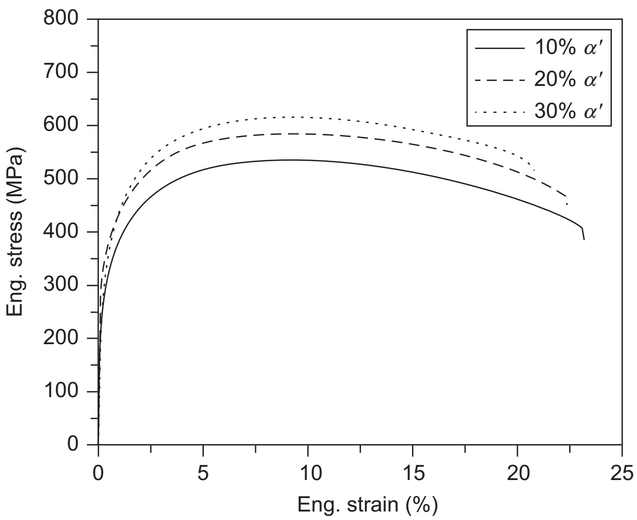

Fig. 12 Stress-strain curves for three DP steels with MVF = 10%, 20%, and 30%.

reported[28] that in martensite structures, C atoms are trapped leading to a distorted bct lattice, i.e., super-saturation with C compared to the bcc ferrite. This leads to the high strength of martensite. In carbon steels with high M_S, tempering of martensite takes place already during cooling and also during storage and testing at RT. Thus, C may precipitate or diffuse toward dislocations and interfaces.

A possible explanation toward this behavior can be given by the specific microstructure. For DP steel having 30% of MVF quite a large number of mobile dislocations at the F–M interfaces could be expected. After deformation, their number even increases. Together with the low volume fraction of ferrite it is probable that the dislocations generated during pre-straining can be blocked very effectively. Thus, the BH values are larger than those with lower MVF.

In this study it was checked whether such a correlation could be found for the DP steels. Comparing the samples with different MVFs it is obvious that in our study such a relation is valid, either for the BH_0 or for the BH_2. The results are presented in Fig. 14. A linear increase of BH_0 and BH_2 with MVF is found for the samples cooled at 10 K/sec. This observation is valid for other cooling rates. However, a final answer, whether this dependence is solely attributable to the amount of martensite or to related changes in the microstructure affecting both R_m and solute C and hence BH_0 and BH_2, is yet to be given.

Conclusions

1. The estimated MVF in the final microstructure is in good quantitative agreement with the experimental data. It is estimated that quenching of specimens from different TFC temperatures results in different DP microstructures containing different MVFs at a given cooling rate. An assessment of the microstructures obtained in the thermo-mechanically processed samples with respect to MVF and cooling rate allows selection of processing parameters required to develop the specified DP microstructures.
2. By making use of the beneficial effect of the TMCP a variation in MVF and cooling rate does not influence the grain size of phases. The dislocation density in ferrite and at F–M interfaces increases with increasing MVF, resulting in a higher fraction of ferrite affected by the martensitic phase transformation and in higher residual stresses introduced into the matrix.

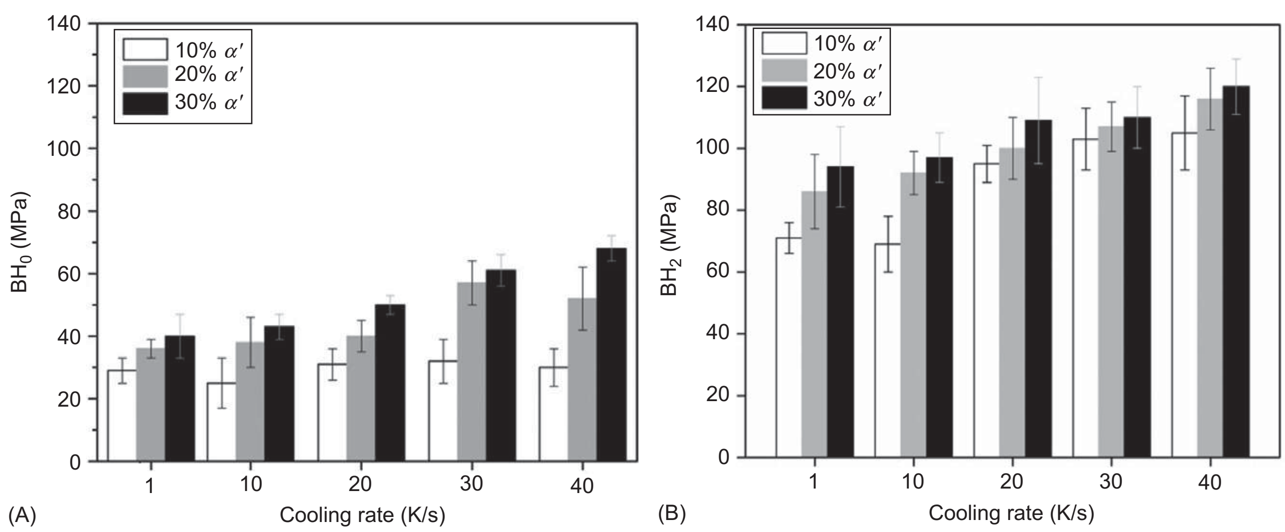

Fig. 13 (a) Dependence of BH_0 on the cooling rate and MVF calculated from difference between $R_{p0.2}$ of the tensile sample and R_e of the respective BH sample and (B) dependence of BH_2 on the cooling rate and MVF.

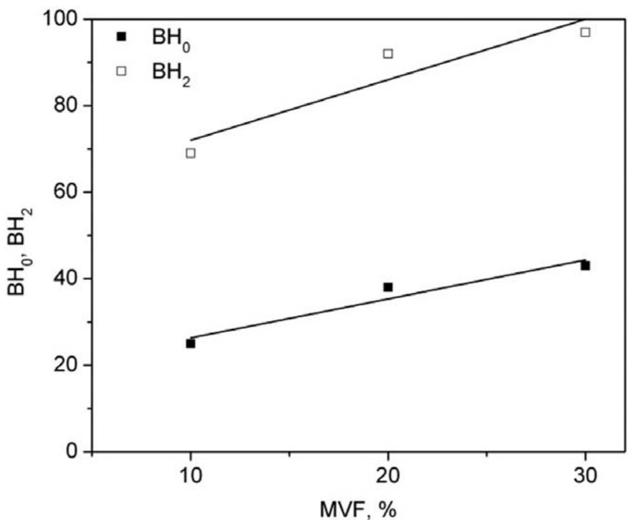

Fig. 14 Relationship between BH values and martensite volume fraction (MVF); cooling rate during the $\gamma \rightarrow \alpha$ transformation was 10 K/sec.

3. An increase in cooling rate after the last deformation step to TFC during the $\gamma \rightarrow \alpha$ transformation results in higher values of strength after TMCP as well as after PS and BH. This is due to an increase in the contribution of thermally activated dislocation processes.
4. The magnitude of mechanical properties after TMCP as well as after PS and BH is affected by the MVF. The increased number of dislocations in the ferrite grains and a higher amount of C atoms at the F–M interfaces is the most reasonable explanation for this observation, leading to a stronger dislocation pinning at large MVFs.

Influence of Chemical Composition

Aim of the study

This part is concerned with a further investigation of hot-rolled DP steels by varying their chemical composition within the standard deviation of the steel plant. The study clarifies how the addition of selected alloying elements (C, Si, Mo, and Nb) affects the microstructure, mechanical properties, and BH behavior through the aforesaid metallurgical factors. The effect of both, alloying elements and TMCP, was studied in detail. Throughout the present work, the influence of alloying elements on phase transformation kinetics had been reviewed.

Investigated materials and alloying concept

For this study, DP steels with chemical composition given in Table 6 was selected as the base steel which was then vacuum-melted in laboratory furnace and appropriately alloyed to obtain different chemical compositions. The selection of the compositions was based upon current researches under consideration of the industrial tolerance extent.

Table 6 Chemical composition of the steels (wt.%); D1: Base steel.

Steel	C	Si	Mo	Nb	Cr	N
D1	0.06	0.10	0.005	0.002	0.60	0.006
D2	**0.09**	0.10	0.005	0.002	0.60	0.006
D3	0.06	**0.25**	0.005	0.002	0.60	0.006
D4	0.06	0.10	**0.25**	0.002	0.60	0.006
D5	0.06	0.10	0.005	**0.04**	0.60	0.006
D6	**0.09**	**0.25**	**0.25**	**0.04**	0.60	0.006
D7	**0.09**	0.10	0.005	**0.04**	0.60	0.006
D8	**0.09**	**0.25**	**0.25**	0.002	0.60	0.006

Thermo-mechanical-controlled process schedule

The TMCP simulation of finishing rolling was applied using flat compression test as described before. The hot deformation parameters of the last three deformation steps were the same as listed in Table 4. The cooling rate had been taken following industrial conditions. The flat compression specimens were cooled after the last deformation step to the fast cooling start temperature (T^{FC}) with a cooling rate of 10 K/sec until the required fraction of ferrite (80%) and austenite (20%) was obtained and then accelerated cooled down to RT with a cooling rate of ~100 K/sec (Fig. 10). As well known, the alloying elements affect the $\gamma \rightarrow \alpha$ transformation temperatures during cooling. Therefore, the T^{FC} temperature in the γ and α regions changes depending on the alloying elements. The T^{FC} temperature for each alloy was determined by means of deformation simulator in.[20]

Results

Microstructure evolution. Microstructure investigation of different alloys reveals that the grain refining effect of alloys D3 and D5 is demonstrative, whereas no refined grains are observed for other alloys. Fig. 15 shows exemplary the microstructure analyses of alloys D1 and D5 after TMCP. In general, the specific characteristics of the microstructural features reveal in the DP alloys with different alloying elements include the following:

a) The volume fraction of ferrite and martensite for all alloys is quite the same 80% and 20%, respectively. The estimation of the MVF is in good agreement with the values calculated from dilatation curves.

b) Large martensite islands can be seen, but they often show dark substructures either within or in their immediate surroundings.

c) The average grain size of ferrite and martensite is finer in alloys containing low C content and addition of Nb or Si. For these alloys the grain refining effect of Nb is more visible than that of Si.

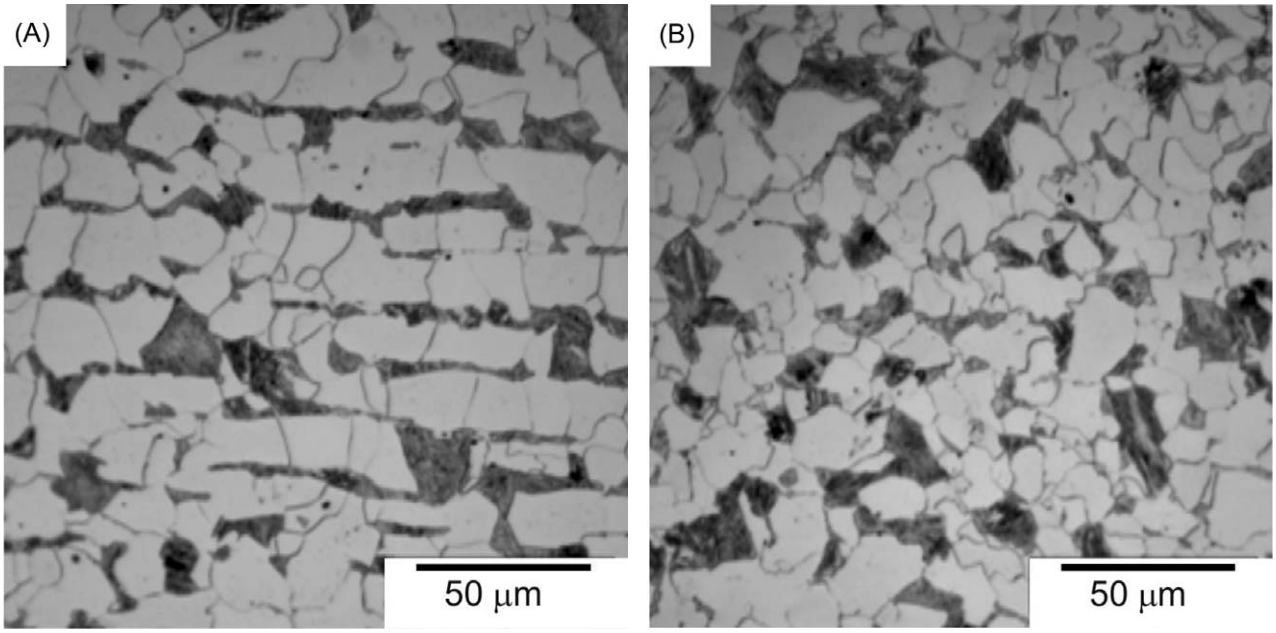

Fig. 15 Microstructure of DP steels showing different grain sizes obtained after TMCP: (A) alloy D1 (B) alloy D5; cooling rate of all samples in first cooling stage = 10 K/sec and in second cooling stage = 100K/sec.

Microstructure evolution after the pre-straining and baking process. TEM analyses were conducted on alloy D1 after TMCP without pre-straining at RT and at T = 170°C and 240°C. Since no significant differences with regard to the morphology and dislocation distribution of ferrite were found in alloy D1 for the given conditions, only TEM images of martensitic phases are evaluated in this section.

Fig. 16 provides overviews of the martensite structure under different conditions. The martensite morphology, showing packets of laths with different orientations, can be seen more clearly from this figure. In addition, a number of very small particles, most probably carbides, within single laths can be detected. TEM experiments on samples baked at temperatures of 170°C/20 min (Fig. 16B) yielded no significant differences to the microstructure of the basic steel (Fig. 16A). Clear changes in the martensite phase were observed after BH simulation at 240°C/20 min. In contrast to the martensite morphology of the sample without baking treatment, strong tendencies for the martensite phase to decompose are found in the baked state at T = 240°C (Fig. 16C). Such decomposition does not necessarily affect the whole structure all at once, but parts of the martensite may keep their initial structure. The formation of carbides can also be observed from Fig. 16C showing small carbide particles inside of the lath martensite. Diffraction patterns confirmed that the carbides are of Fe_3C type (Fig. 16D). They indicate possible martensite aging as a result of the baking process.

Mechanical properties. Mechanical properties corresponding to all of the investigated alloys after TMCP are displayed in Fig. 17. Stress–strain curves (Fig. 18) have been printed to show changes in the yielding behavior of the different alloys. Evaluating them, a pronounced influence of the chemical compositions on mechanical properties is obvious. For all alloys investigated, the amount of martensite was kept constant. Among the alloys with higher C content, D8 indicates the highest level of ultimate tensile (R_m) and yield strength ($R_{p0.2}$) combined with a narrowed total elongation (TEl) with R_m = (728 ± 18) MPa, $R_{p0.2}$ = (469 ± 14) MPa, and TEl = (14 ± 0.5)%. The comparable values of the basic alloy D1 are R_m = (607 ± 15) MPa, $R_{p0.2}$ = (375 ± 12) MPa, and TEl = (20.5 ± 1)%. Alloy D6 with the same alloying elements as D8 but an increased Nb content demonstrates nearly the same level of strength as alloy D8 but a slightly higher ductility. Alloys D2 (with higher C content) and D7 (with higher C and Nb content) also show high strength levels but lower than the ones of the alloys D6 and D8. D2 and D7 reveal a higher ductility than the alloys D6 and D8.

As expected, the strengthening effect of DP steels generally increases with increasing C content. Thus, C is the most effective element in increasing the strength of low carbon steels under the chosen conditions—change of weight-% within the standard tolerance range—and reveals an essentially strong effect in DP steels together with increasing Si, Mo, and Nb approaching to 0.25%, 0.25%, and 0.04%, respectively.

Among alloys without C addition, D3 with the highest Si content indicates nearly the same strength and ductility level as D4 with the highest Mo content. The tensile curves of D3 and D4 show that very similar properties

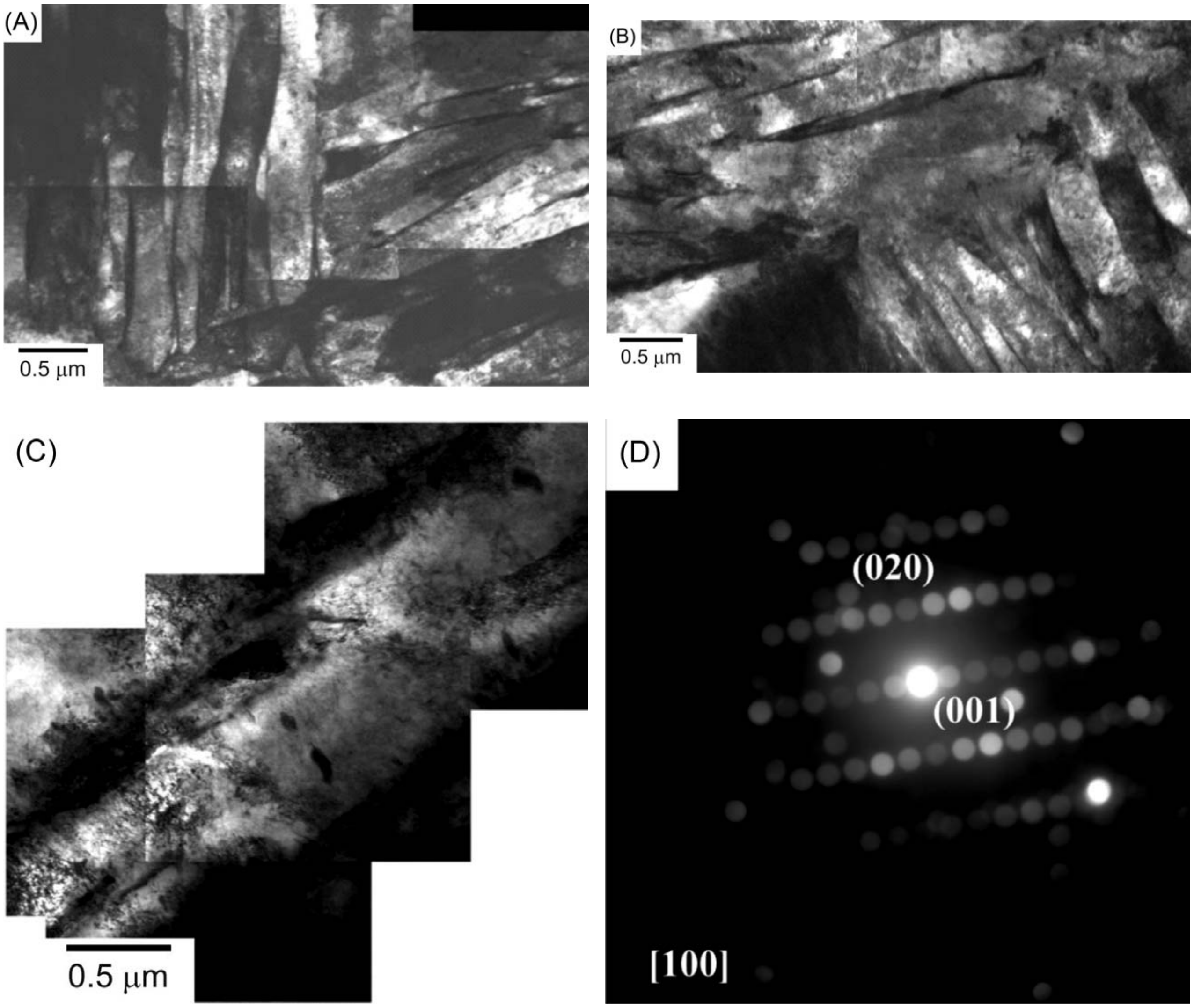

Fig. 16 TEM micrographs showing lath martensite obtained from steel D1 after TMCP (A) without PS and BH; (B) after PS to 2% and BH at 170°C for 20 min; (C) after PS to 2% and BH at 240°C for 20 min; and (D) diffraction pattern from cementite within single laths.

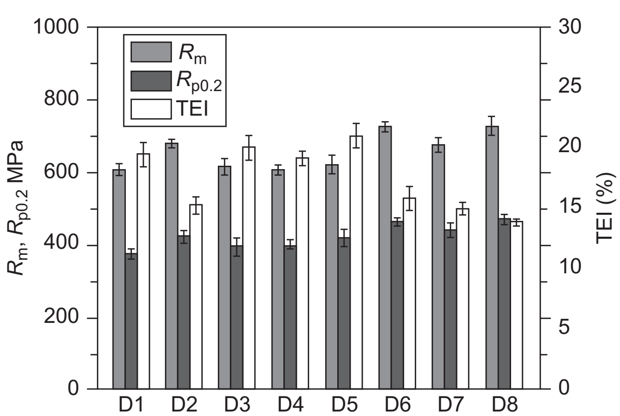

Fig. 17 Tensile strength (R_m), yield strength ($R_{p0.2}$), and total elongation (TEl) dependent on the chemical compositions (D1–D8) of the DP steel produced by TMCP.

can be achieved for these steels. R_m close to 620 MPa was found for both alloys. No significant differences were observed for the elongation properties. Among the alloys without any C addition, the highest values of R_m and $R_{p0.2}$ were observed in Nb-microalloyed steel (D5). In all cases, addition of Nb seems to improve the ductility.

Bake hardening behavior. In Fig. 19, the BH_0, BH_2, and BH_5 values for different alloys with respect to the BH are shown. These figures yield some clues about the possible mechanisms contributing to the BH behavior of DP steels. Considering the BH behavior under amounts of pre-straining a higher BH level is found for the pre-strained steels (BH_2 and BH_5) compared to non-pre-strained ones (BH_0). The greatest level of BH is obvious for steels pre-strained with 5% (BH_5). Comparing the different baking temperatures (170°C and 240), clearly higher BH values

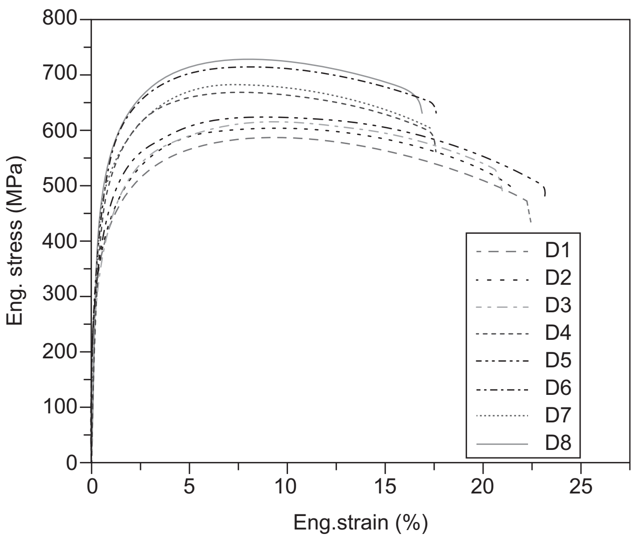

Fig. 18 Engineering stress–strain curves of DP steels containing different amount of alloying elements. The DP steels having 20% of MVF are obtained by TMCP.

are found for 240°C. The difference between 170°C and 240°C is more demonstrative for BH_2 and BH_5 at the pre-strained condition. It is of less evidence for BH_0 when samples are not pre-strained.

Comparing the alloys with respect to BH values, D2 and D8 demonstrate the highest BH values with a pre-strain of 2% and 5% (Fig. 19). For alloy D6 a high level of BH, but slightly lower than for D2 and D8, is found under these conditions, while D6 without pre-straining shows a higher BH level than alloy D2 under comparable conditions. Among the steels with the highest C content, D8 exhibits the largest BH, while D7 reveals the smallest one.

In general, for alloys with a low C content a lower R_m level after PS and BH is found compared to alloys with higher C content (Figs. 17 and 18). Among these alloys, D5 demonstrates the highest strength level after PS and BH, whereas D1 (basic steel) indicates the lowest strength level.

Evaluating the low C content alloys with respect to BH behavior, a significantly higher BH level is obtained for

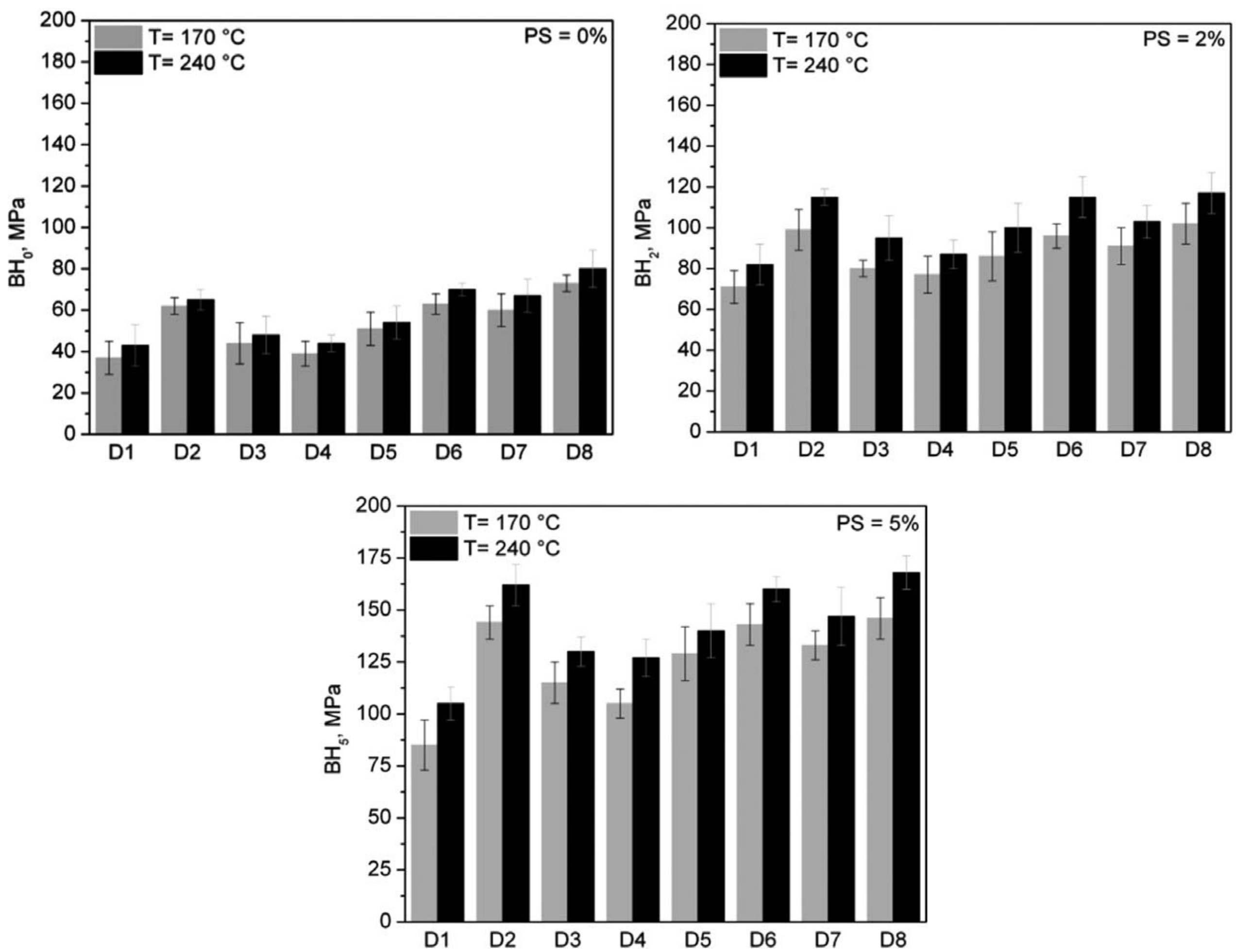

Fig. 19 Influence of the chemical composition (D1–D8) in the DP steel on (A) BH_0, calculated from the difference between $R_{p0.2}$ of the tensile sample and R_e of the respective BH sample, (B) BH_2, and (C) BH_5.

the Nb-microalloyed steel D5 after PS and BH under different conditions (Fig. 19). D5 presents a comparable BH level to D7 with a high C content. D3 also shows high BH values after PS and BH which are still lower than those of D5. For alloys D1 and D4 nearly the same BH level is observed in all conditions of pre-straining and baking. However, D4 has a higher BH level than D1.

Discussion

Influence of alloying elements on the bake hardening behavior. Two main groups may be defined from the results of mechanical properties after PS and BH process: Alloys containing low C content (D1, D3, D4, and D5) and alloys containing high C content (D2, D6, D7, and D8).

First, the BH behavior of low C content alloys is discussed. The results of BH clearly show that the BH level of D3 and D5 is the highest among the four low C content alloys (Fig. 19). Alloy D3 with an increased Si content as a solid solution strengthening element shows intermediate values of strength increase after PS and BH. For D5, somewhat higher BH values are obtained. The high BH level of alloys D3 and D5 will be due to finer grain sizes. Grain refining of ferrite by Nb addition in solid solution[29] and by increasing Si[21] has already been reported.

Grain size is reported to be a long range barrier to the dislocation motion in bcc lattice and hence affects the BH.[10,21] The reason why bake hardenability depends on grain size is not clear yet, but it is inferred that the influence of dissolved carbon on the bake hardenability differs depending on the location of C. Different effects of dissolved C were reported on the bake hardenability depending on its location, at grain boundary and inside grains.[30] For a given carbon content, the BH increases with a decrease in the grain size and the dependence on grain size increases with an increase in solute carbon. While the explanation of this effect is not complete, data suggest that free C located near grain boundaries has a more profound influence on the strength than free C located within the grain interior.[31]

Comparing D4 with D3 and D5, reduced bake hardenability is found for D4. As mentioned earlier, the presence of the stronger carbide-forming elements, like Mo in steel D4, may lead to the formation of its carbides (Mo_2C) in addition to Fe_3C. The presence of Mo_2C is confirmed by EDX investigations.[20] Therewith, the amount of solute C decreases because it is tied up in the alloy by carbide particles. Compared to D3 and D5 this may cause the reduction of bake hardenability.

Evaluating the BH behaviors of alloys with higher C content shows a significant higher BH level for these alloys compared to the alloys with a low C content. Higher C content in the alloys is associated with a larger magnitude of solute C. To maximize the strength increase associated with BH, it is necessary to have as much free carbon as possible. The classical BH mechanism is related to pinning of mobile dislocations by C atoms. The main part of this C is found in the C-rich phases like martensite, retained austenite and bainite, of course. But there are still reasons to assume that some C is dissolved in ferrite as well, resulting from the fast cooling after intercritical annealing or after TMCP, when the solubility for C in ferrite is higher than at RT. Thus, some excess C can be kept in solid solution. Nevertheless, significantly larger amounts of C in the ferrite phase are not likely for multiphase steels, since their alloying with manganese promotes the formation of austenite during the $\gamma \rightarrow \alpha$ transformation, which in turn narrows the ferrite phase field.

Evaluating the BH behavior of alloys D6 and D8, a comparable high BH level is obtained for both alloys. Both alloys contain the highest amounts of C, Si, and Mo, while alloy D6 is additionally micro-alloyed with Nb. As discussed before, the grain size becomes finer when Si[29] and Nb[30] contents are increased. On the other hand, no grain refining in high C content alloys is observed by increasing Si and Nb. Thus, the high BH behavior of high C alloys can be well explained in terms of the changes in solute C.

Among the alloys with increased C content, D7 shows the lowest values of BH. This could be due to the formation of Nb (C, N) which could lead to a reduction of solute C. Nb forms Nb (C, N) which may tie up the solute C contributing to BH.

Influence of the pre-straining and temperature on the bake hardening behavior. Stress vs. strain curves for alloys D1 and D6 (Figs. 20 and 21) have been produced to show changes in the yielding behavior with respect to PS and baking temperature.

This yields some clues about the possible mechanisms contributing to the BH behavior of DP steels. Alloys D3,

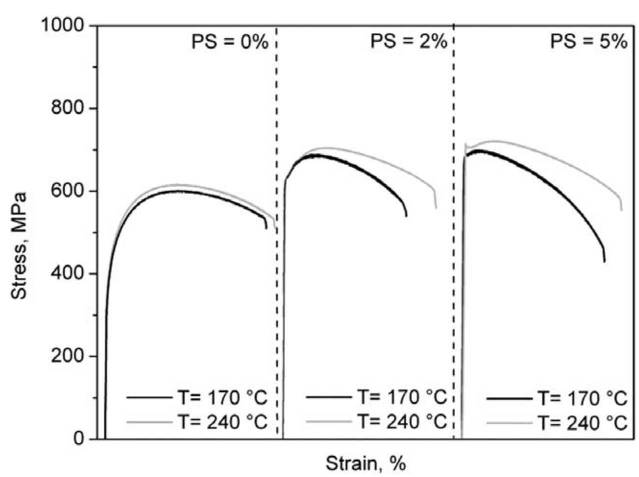

Fig. 20 Stress–strain curves for alloy D1 after PS and BH.

D4, and D5 with a low C content show quite similar stress curves. Without PS no or just minimal deflection is visible, indicating aging effects. This is accompanied with very low BH_0 indices. A more distinct response to BH can be observed at 2% and 5% pre-straining at which a real yield point phenomenon occurs. The occurrence of a yield point indicates that a dislocation pinning mechanism becomes active.

Significant changes in the yielding characteristics of the tensile curves with increasing temperatures are found. For all four alloys distinct aging is observed at 170°C. At the BH temperature of 240°C the yielding characteristics become very pronounced showing strong yield drops up to yield point elongations. It is to note that yielding is discontinuous at all imposed pre-strains and temperatures, excepted for PS = 5% and T = 170°C. For this condition, continuous yielding appears for all alloys investigated. The yield point elongation becomes more pronounced at a temperature of 240°C. In general, the formation of Lueder's bands is more pronounced at elevated temperatures and PS. The continuous increase of BH values at 240°C can be attributed to the faster diffusion of C atoms toward the dislocation lines. The highest BH_2 and BH_5 values are observed for this high temperature. This indicates that additional processes start to become involved at elevated temperatures. Some authors report tempering mechanisms of martensite at temperatures between 200°C and 250°C in DP steels.[30] In martensite structures, C atoms are trapped leading to a distorted bct (body-centered tetragonal) lattice, i.e., super-saturation with C compared to the bcc ferrite. In C steels with high amount of martensite, tempering of martensite takes place already during cooling and also during storage and testing at RT. Thus, C may precipitate or diffuse toward dislocations and interfaces.

Fig. 21 presents the general characteristics of the stress–strain curves of D6 with the highest C content level. Evaluating the tensile curves pre-strained to 5% and baked at 170°C and 240°C, respectively, a continuous decrease of stress after reaching the upper yield point is observed. Obviously, baking increases the R_e close to the R_m, i.e., after the onset of yielding the sample instantly starts to neck. In this case, the degree of PS approaches the uniform elongation of the particular alloy.

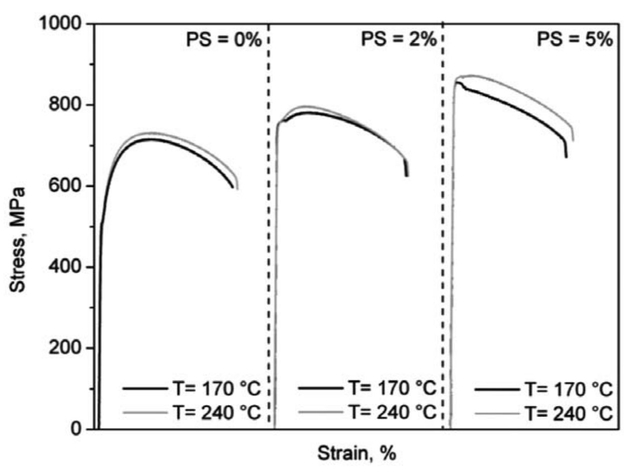

Fig. 21 Stress–strain curves for alloy D6 after PS and BH.

A first indication of aging can be seen in the curves after BH simulation at 240°C. Especially for steel D8 this leads to a BH as high as 120 MPa at PS = 2%. The occurrence of a distinct yield point allows the conclusion that at 240°C baking temperature a C-related mechanism has a distinct influence on the BH behavior. Such a mechanism is probably present at lower temperatures as well, since cooling of the alloys after the last deformation step during TMCP and accelerated cooling from fast cooling start temperature (T^{FC}) should provide a certain amount of C in solid solution, but it might be overshadowed by a strong tempering phenomena taking place in these alloys. It is reported that such C is present at sites where it is hindered to contribute to baking at low temperatures.[30] Possible sites where C resides are regions in the vicinity of the ferrite and martensite interfaces with a high density of geometrically necessary dislocations.

It should be noted that these alloys show higher yield points after PS and BH treatment than those alloys containing low C content. It can be conjectured that this effect is attributable to a large amount of solute C present in such steels due to higher C content. It is probable that the dislocations generated during pre-straining can be blocked very effectively.

Two main conclusions from these data on the response to BH are: 1) the contribution of BH increases with an increase in pre-strain and 2) the BH sensitivity of the baked alloys at 240°C is much higher than that one at 170°C.

The transformation of austenite to martensite leads to a volume increase which has to be compensated by the generation of dislocations in the surrounding ferrite. During the transformation process ferrite to martensite compressive stresses are introduced in the surrounding ferrite causing an increase of yield strength. As mentioned earlier, TEM analysis on the alloy D1, pre-strained to 2% and without baking yield, show a negligible higher dislocation density compared to the condition before pre-straining. Moreover, no significant differences are observed after BH at 170°C/20 min to the microstructure (martensite phase) of the basic alloy (Fig. 16A–B). A distinct tempering of martensite takes place after BH simulation at 240°C/20 min, showing a clear decomposition of the martensite structure and the formation of carbides (Fig. 16C). Diffraction pattern confirms the Fe_3C carbides (Fig. 16D). This can lead to a high BH level at this temperature.

Conclusions

1. Among the four alloying elements, increasing C and Mo does not influence the ferrite and martensite grain

sizes. Increasing Nb and Si refines the final microstructure of DP steel when they are added into steels individually. The effect is more pronounced for the addition of Nb. No grain refining effect is observed for the steel by addition of Nb together with increasing C.

2. Being with all alloying elements at the upper limit increases the strength of DP steels. C is the most important element influencing the mechanical properties of DP steels. Nb is more effective than Mo and Si in increasing the strength without deteriorating the ductility. Since much of the Nb effectiveness in steels with high C content results from complex two-, three-, and four-way alloy interactions, Nb alloying does not cause an increase in the strength of steels unless Mo and Si are added additionally. It thus allows working with an alloying concept to reach a specified strength level.
3. Variation of alloying elements as well as pre-strains and baking temperatures indicate a pronounced influence on the BH behavior of DP steels as follows:
   a. C is the most important element affecting BH behavior. The steels investigated having high C content exhibit a high BH potential.
   b. Increasing Si and Nb improves the BH behavior in steels without rising C, where Nb is more effective than Si in increasing the BH behavior. This is due to grain refining effect of both elements. Increasing the C content of DP steels from (0.06–0.09)% strongly reduces the effectiveness of Nb alloying on the bake hardenability. The formation of Nb (C, N) precipitations is used to scavenge solute C from the ferrite matrix in the DP steel. Partial redissolution of Nb (C, N) during the roughing rolling can be used to liberate C for the BH effect.
   c. For Mo increasing in steels without C rising a slightly rise of the BH is observed. Increasing Mo together with other alloying elements shows a rising effect on BH.
   d. For all alloys, a strength increment is observed without PS and with pre-strains at 2% and 5% after the bake hardening heat treatment. This increment is significantly high for the alloys pre-strained at 5% and baked at 170°C and 240°C for 20 min where for the baking temperature of 240°C a slightly higher BH level is found.

These investigations on process temperature and strain distribution as well as the cooling conditions and usual changes in the alloying elements show the variety of the BH values within one quality of steel. Moreover, knowing the interdependencies they give the possibility to react on these deviations in an early state of production and level the process for a proper design of the final material's properties.

CHAPTER 2: INFLUENCE OF THE LOCAL USE OF AGING AND BAKE HARDENING EFFECTS IN DP STEELS

As the deformation and heat-loading shaping and joining a part is not always performed globally the local BH in multiphase steels comes into the focus of investigation automatically. The goals of this subpart are to analyze the effect of the local influence of BH treatment and to suggest solutions for processes being able to induce a locally restricted aging effect. The potential and efficiency of different methods and procedures of local forming and local heat-treating processes are studied.

Materials and Experimental Methods

Though different types of multiphase steel were investigated,[36] in this chapter only results of the DP steel are introduced. The chemical composition of the DP discussed is the same as used before and listed in Table 1.

Two methods were investigated to achieve local strengthening. In the first case, the sheet is deformed partially and afterward heated globally—following the shaping procedure with a final baking process. Local strengthening was induced with an embossing machine (Fig. 22) providing the option of rolling straight or curved contours into the sheet to affect the behavior of structural elements or components within a definite size and shape by means of a local deformation, causing an increase in dislocation density.

The sheet is clamped onto a plate, which is movable in longitudinal direction. An arm with a roller, moveable around its axis and in transversal direction, is mounted above this plate. The arm transmits the hydraulic forces to a deformation roll. The embossing depth, the hydraulic

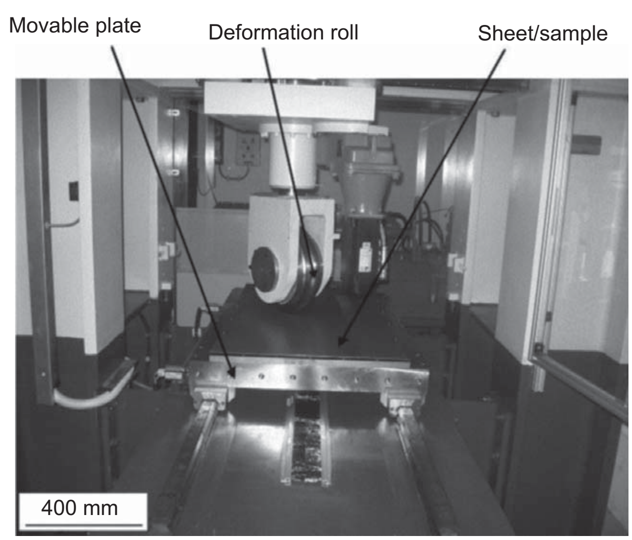

Fig. 22 Embossing machine for local deformation.

Table 7 Specific data of the embossing machine.

Max. sheet area (mm)	1100 × 600
Min./Max. force of roll (kN)	1–300
Max. deformation speed (mm/sec)	18
Roll diameter (mm)	350

force, as well as the movement of the arm are computer-controlled. Table 7 lists some specific data of the machine.

The production of a defined deformation is made possible by setting the embossing depth. Table 8 shows the specified embossing depths and their corresponding true strains for the selected materials. The sheets were deformed locally with the defined strain and then subsequently aged at different temperatures (100°C, 170°C, and 240°C) for a holding time of 20 min following the conditions of the standard painting processes in the automotive industry.

A second method was to deform the sheet globally with defined degrees of deformation and to heat treat it locally with a laser. The local laser heat treating (LHT) is accomplished using a laser-remote-welding equipment. To understand the dependencies, samples of size 500 mm × 80 mm × 2.0 mm were cold rolled homogeneously with different thickness reductions ($\varepsilon = 2\%$, 5%, or 10%) before LHT. For the introduced investigations, the laser speed was fixed to 5.0 m/min. The LHT for the DP steels was conducted with laser powers between 600 W and 2800 W to investigate the influence of the energy input.

Tensile tests performed on the samples followed LHT and LD in tension direction using an Instron 250 kN machine. For an initial overview of the expected effects, hardness measurements had been carried out by Omnimet MHT Buehler using a Vickers micro-indenter according to the standard DIN 50133.

Results

Local deformation by embossing

Fig. 23 shows an example of the estimated hardness profiles along the deformation line and in the nondeformed areas as a function of different aging temperatures. The results show the hardness obtained for the DP steel with an embossing depth of 0.1 mm (true strain = 0.05), following the standards for BH material at a holding time of 20 min. The hardness values were determined both in the locally deformed region and in the basic sheet. The figure shows the expected increase in hardness of about 20 HV0.1 following local deformation. Compared to the initial state, the heat treatment at 100°C produces no relevant change in the hardness values. At 170°C, an increase in hardness

Table 8 Embossing depths and corresponding true strains.

Embossing depth (mm)	0.05	0.35	0.50	1.00
True strain φ	0.05	0.02	0.10	0.15

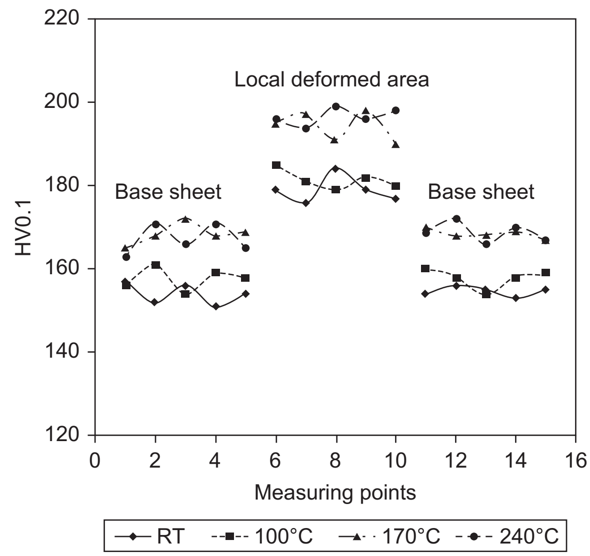

Fig. 23 Hardness values of DP steel in the bottom of the locally deformed region (middle) and in the base material (left and right) followed by an ageing treatment for 20 min. Deformation in the bottom of the embossing line: $\varphi = 0.05$.

of about 15 HV0.1 in the locally deformed region was obtained. In addition to this, there was also an increase of hardness in the basic metal at this temperature. A further increase in the aging temperature up to 240°C did not show any change in the hardness value compared to that at 170°C.

Fig. 24 presents the averaged values of the mechanical properties for the DP steel with respect to the pre-strain (φ) and the aging temperature. Three samples per test were used. The yield and tensile strength values rose steeply as the pre-strain increased to $\varphi = 0.1$, while the total elongation decreased with increasing pre-strain as expected. This observation is valid for all aging conditions, without aging and with aging at different temperatures. The influence of the aging temperature in general can be described as to rapidly increase the strength values at higher aging temperatures (170°C and 240°C), while at lower aging temperature (100°C) only a slight increase, which is more significant for the tensile strength at higher pre-strain rates, can be observed. The total elongation decreased with increasing aging temperatures.

Local laser heat treatment (LHT)

The hardness distribution for the DP steel following a global deformation by cold rolling with $\varepsilon = 5\%$ and a laser treatment with different laser powers at constant laser speed is presented in Fig. 25. After cold rolling the measured hardness was (235 ± 8) HV0.5. For a laser power of 0.6 kW and 1.4 kW, a minor increase in hardness could be observed. For a laser power of 2.2 kW, the highest hardness increase of (135 ± 9) HV0.5 and, consequently, local

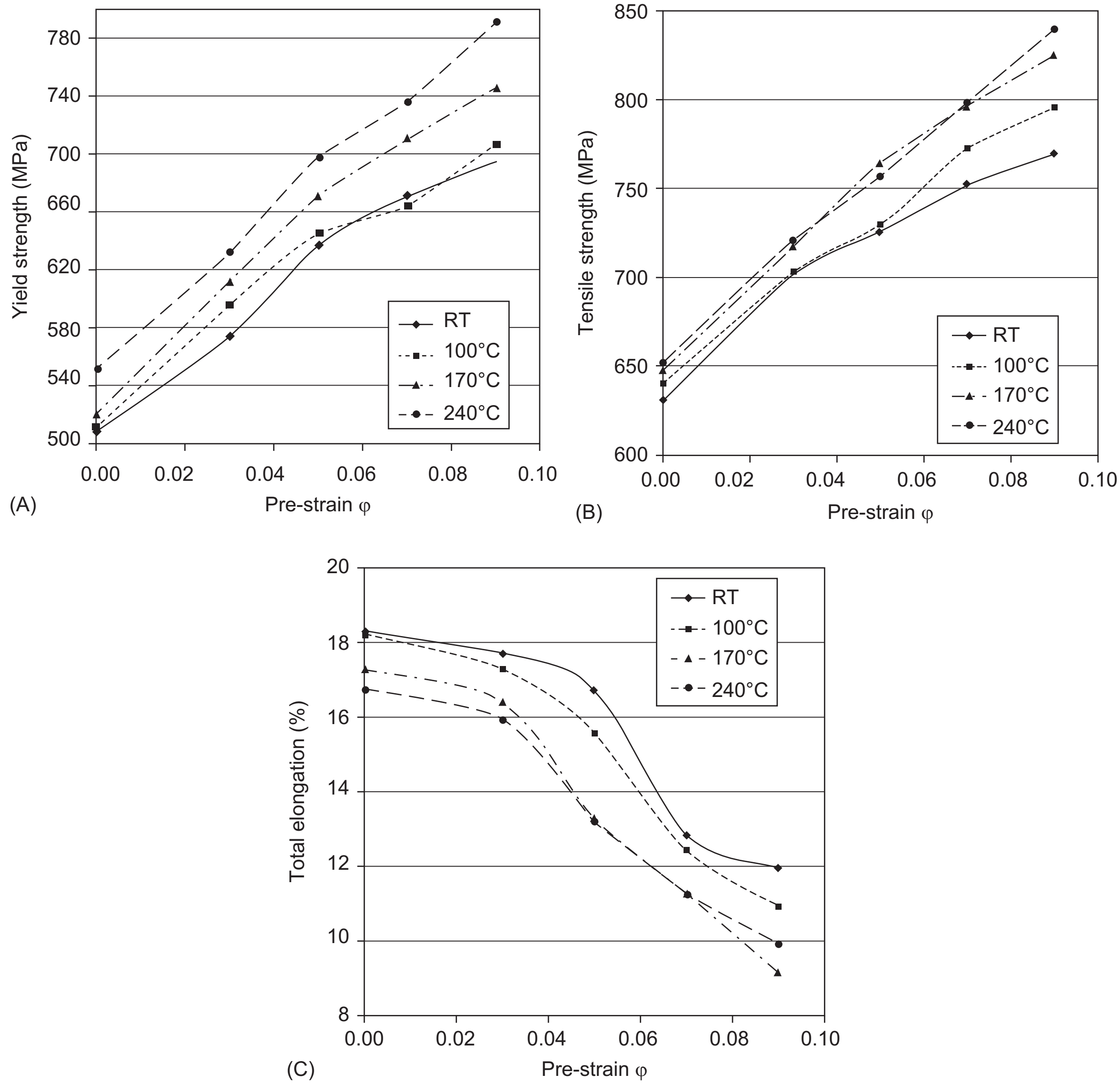

Fig. 24 Mechanical properties of the DP-steel after local embossing with different degrees of pre-strain (φ) and global ageing treatment at different temperatures, holding time: 20 min.

strengthening was measured. For 2.8kW, a significant hardness increase of (120 ± 11) HV0.5 could be observed. Compared to 2.2 kW, the hardness distribution decreased in the middle of laser-treated region. This is due to an overheating effect. The reason for the enhancement of the hardness values of samples, which were treated with lower laser powers, compared to the non-laser-treated condition, could be contributed to the bake hardening effect. The significantly higher hardness values at higher powers are due to affecting of the microstructure. The higher heat-treating temperatures lead to the formation of a large number of martensite and bainite with local plastic zones, having a high dislocation density, in polygonal ferrite and local strengthening due to phase transformation.[37,38] This conclusion could be confirmed by their microstructures.[36]

Additionally, the hardness HV0.5 over the cross-section of the samples in the laser track zone was measured in order to analyze the difference in strengthening locally. As an example, Fig. 26 shows the hardness distribution and microstructure in such a cross-section for the DP steel with a reduction of $\varepsilon = 5\%$. The sheet was laser-treated with a power of 2 kW at a speed of 5 m/min leading to a maximum temperature of ~1000°C at the surface. The change of hardness correlates with the change in the microstructure in the heat-influenced zone. The maximum

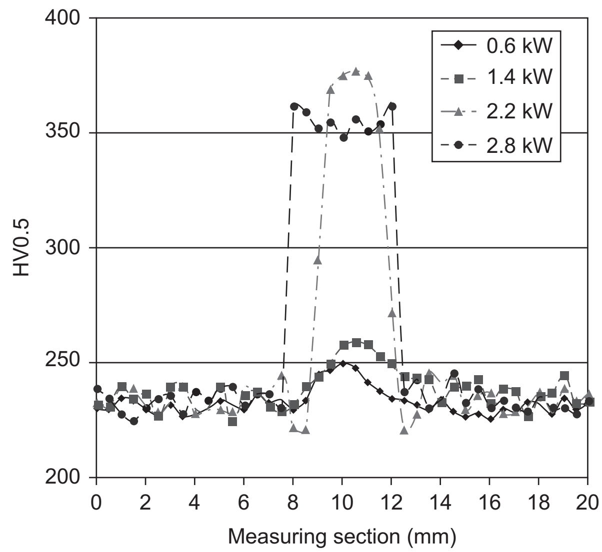

Fig. 25 Hardness distribution of the DP steel sheet, globally cold rolled (ε = 5%), after laser treatment with different laser powers, laser speed: 5 m/min.

of hardness of about (345 ± 6) HV0.5 was observed in the top line of the sheet (position 1), which consists of a martensite and bainite phase. The hardness decreases in the transition zone (position 2) about 50 HV0.5. In this zone a lath martensite, bainite, and ferrite phase could be detected. In position 3, a hardness level of (256 ± 5) HV0.5 was measured, showing a large number of ferrite, less bainite, and a small number of martensite. The hardness values in position 4 are identical to that one in the base metal. Thus, this demonstrates clearly the gradient

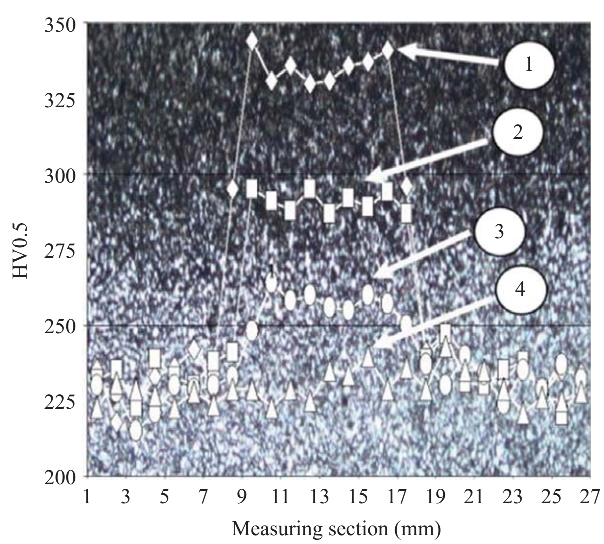

Fig. 26 Micro hardness profile (over the cross-sectional plane) of the laser-treated DP steel (ε = 5%), laser power: 2 kW, laser speed: 5 m/min.

effect of a laser treatment and the possibility to influence microstructure more or less strongly, depending on the parameters used.

To investigate the changes in the mechanical properties by LHT and to show the stiffening effect of this treatment, tensile tests were carried out additionally. For detecting the local flow behavior the optical measuring system ARAMIS, Gom, was used. Fig. 27 shows the tensile sample, cold rolled with a reduction of ε = 5%, with the three laser-treated areas (2.8 kW, 5 m/min) and its strain distribution before necking. The intensive hardening effect in the heat-treated zone is significant. The local laser treatment leads to a strengthening of the steel and therewith reduced deformation in these areas. Necking always occurs in the nontreated areas! So partial heat treatment—e.g., with a laser—can be used for local reinforcement.

Long-time stability of the local treatment

Structural components of multiphase steels normally show a natural aging behavior. The questions to be answered now concern both the stability of the local aging by means of this natural aging process and the long-term behavior of the nontreated regions. In particular, it has to be examined to what extent the local influences can be stabilized and used to improve the material properties of the entire construction unit.

In industrial condition, steel manufacturer supply bake hardening grade steel to the automotive manufacturer with a shelf life of minimum 3 months (~12 weeks). In order to simulate the industrial shelf life condition as well as to save time, samples were aged at an elevated temperature

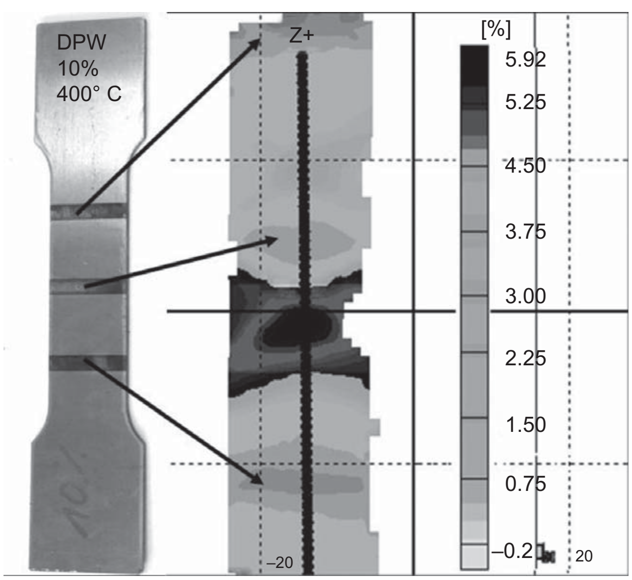

Fig. 27 Local strengthening by laser in DP steel, cold rolled with ε = 5%, laser treated with a laser power of 2.8 kW and a laser speed of 5 m/min.

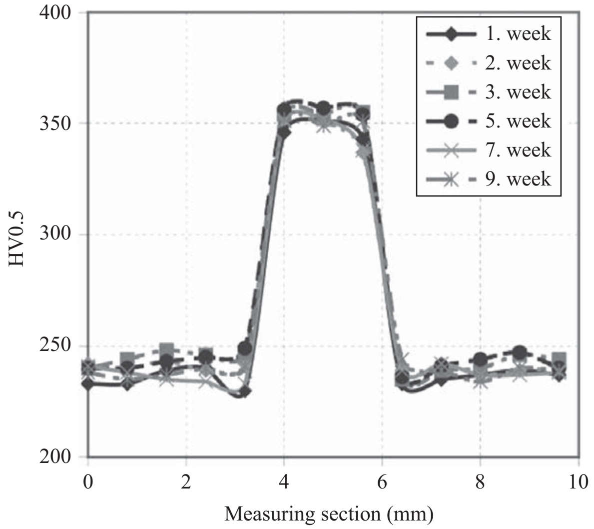

Fig. 28 Long-term aging of laser-treated DP steel, cold rolled (ε = 10%), and locally laser treated, laser power: 2.2 kW and laser speed: 5 m/min.

of 70°C for up to 9 weeks to accelerate aging at shorter times, converging to industrial conditions. Several studies show the aging effect of multiphase steels depending on the aging time (t) and temperature (T).[39,40]

As an example, Fig. 28 shows the measured hardness HV0.5 for the DP steel, pre-strained by cold rolling with ε = 10%, partially laser treated with a laser power of 2.2 kW and a laser speed of 5 m/min. The samples were aged for Cottrell atmosphere formation at an elevated temperature of 70°C for up to 9 weeks. A minor change of hardness in the heat-treated zone could be observed. In the nonaffected zone, an increase about only 10 HV0.5 could be measured over time, a measure that is expected because of the pre-straining but is far away from the effect induced by the laser treatment. So, the locally adjusted strengthening effect can be stated to be stable! This aging stability could also be established for DP 600 subject to changing of pre-strain and laser power.

Conclusions

1. An increase in hardness and strengthening following a local deformation through embossing result from work hardening. An aging treatment at temperatures between 170°C and 240°C lead to a further enhancement of the mechanical properties.
2. The local heat treatment using laser beam leads to a significant increase of the mechanical properties (hardness and strength) and to a local strengthening of the material.
3. The reason for the enhancement of the mechanical properties of samples, which were treated with LHT at lower temperatures, compared to the non-laser-treated condition, could be contributed to the bake hardening effect. The presence of high volume fractions of martensite and bainite at even higher temperatures (above the annealing temperature) lead to the formation of local plastic zones, having a high dislocation density, in polygonal ferrite and local strengthening due to phase transformation.
4. The influence of over aging in locally strengthened regions was investigated. The results show that for the parameters considered, the locally adjusted strengthening effect is stable.
5. Local strengthening of multiphase steels opens up new fields of application for the local use of aging and represents a modification of an existing process. The opportunity is provided for influencing only those regions of interest and thereby unlocking the potential for energy-saving.

CHAPTER 3: INFLUENCE OF LOAD PATHS AND BAKE HARDENING CONDITIONS ON THE MECHANICAL PROPERTIES OF DUAL-PHASE STEEL

The BH of a material is characterized by standards, using the tension test, a pre-strain of 2% and annealing conditions of 170°C/20 min. As could be shown earlier,[41] the BH value is not constant but besides other parameters a function of the load path. So, the aim of this subpart is to give information about the deformation behavior of multiphase steels under different load conditions.

The automotive industry in particular is responding with growing interest in using the aging effect. The aging procedure is based on the effect that in a shaped construction unit, dislocations are pinned by carbon and/or nitrogen atoms in solid solution following thermal treatment. This lack of dislocation mobility increases the unit's strength. In the automotive production process, the thermal treatment is carried out at the end of the production line by means of the paint baking process. The combination of pre-strain, baking temperature, and aging time results in an increase in the material's yield strength. If it is possible to compute this increase in yield strength, then a construction unit can be optimized for beneficial weight or crash characteristics. This requires the examination of the material-specific shaping and heat-treating processes since the BH effect in complex-shaped construction units (e.g., tailored blanks, B-pillar, or a crash protection unit) results in locally different strength behaviors following the thermal treatment of the units. For this reason, the mechanical-technological properties were analyzed particularly with regard to temperature, holding time and pre-strain. Under these aspects two basic questions have to be answered:

Table 9 The change in length and the corresponding degree of pre-strain for the DP steel for different types of load path.

Type of load path	Change in length (mm)	Degree of pre-strain ε (%)
Uniaxial	—	2, 5, 10
Biaxial	5, 10, 15, 20	2, 3, 4, 5, 7
Plane strain	6, 7, 8, 9, 9.5	0.50, 0.75, 1.50, 2.25, 4.00

Where is the maximum increase of yield strength and tensile strength at a given pre-strain and what influence does the load path (kind of pre-strain) have?

When does the yield strength reach a maximum as a function of pre-strain, temperature, and holding time and can this effect be computed?

Considering an actual shaped construction unit it becomes clear that the deformation behavior of the material cannot be completely described by using uniaxial tension tests.[41] On this basis, the influence of the type of pre-strain on the aging effect was investigated.

Materials and Experimental Methods

The DP steel used was the same as shown in Table 1. In order to generate different strain paths to cover the Forming Limit Diagram (FLD), uniaxial, biaxial, and plane strain conditions were used.

Uniaxial tests were conducted using a universal tensile machine UTS 250 kN. The samples were stretched to specified pre-strains (Table 9) and subsequently aged for 20 min at 100°C, 170°C, and 240°C. Finally, the tensile test was performed on the samples to determine the mechanical properties.

Considering the different choices to generate biaxial stress, a Marciniak forming tool (Fig. 29) with a diameter of 250 mm was chosen, mounted on a 2500 kN hydraulic press. Thus, it is possible to produce biaxial pre-strained specimens of large dimension. For biaxial pre-straining, the specimens were cut into 400 × 400 mm square sheets. The specimens are strained by a flat-bottomed cylindrical punch. Between the punch and the specimen is a steel driver with a central hole. By means of this, frictionless deformation of the sheet in the hollow area takes place. Using a large width specimen, the external forces acting on the sample are greater than the internal forces, resulting in biaxial condition.[41]

To control the plastic deformation and adjustment, the noncontact photogrammetrical measuring system ARGUS (gom) was used. This system is well suited for measuring three-dimensional deformation and strain distributions in components subject to static or dynamic load, especially for small ranges of elongation.

The 240 mm long tensile specimens were taken from the stretched drawn regions of the sheets (Fig. 29). The tensile specimens were subsequently age treated under the same conditions as the uniaxial pre-strained specimens. Finally, the tensile tests were performed on the samples to determine the mechanical properties of the pre-strained areas and thus ascertain the effect of pre-straining on the strengthening.

For the plane strain test, a special specimen geometry and tool had been developed by Vegter,[42] see Fig. 30. This application allows performing plane strain conditions using a universal testing machine. One of the difficulties is to achieve a uniform plane strain state across the specimen, as is shown in Fig. 31. Due to the existence of a uniaxial stress state at the edges of the specimen,

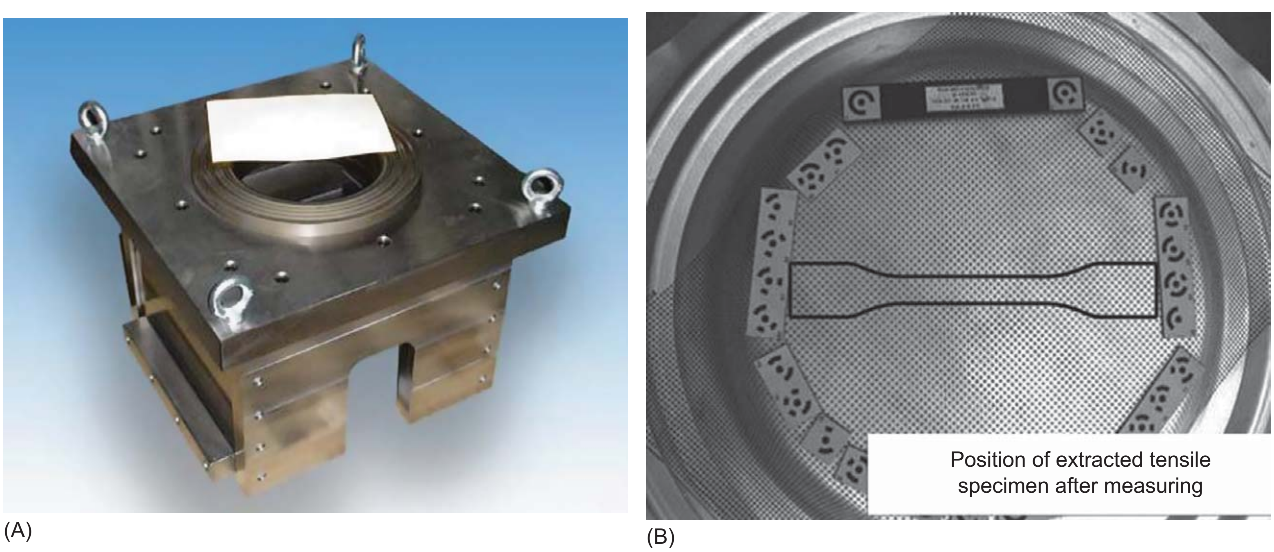

Fig. 29 Marciniak tool to produce the biaxial strain condition. The specimens were cut into 400 × 400 mm sheets. For illustration a white A4-sheet is placed on the tool (left side).

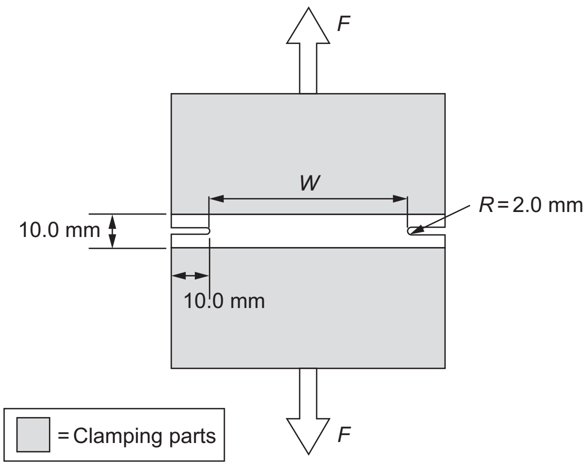

Fig. 30 Specimen for the plane strain test according to Vegter.[39] Specimen size: 300 mm × 300 mm.

deviations from a plane strain state always occur in this type of specimen. A correction has to be made to subtract this edge effect from the test results. A simple geometric average stress–strain curve over the whole width of the specimen would result in a 2% lower value of the plane strain stress. Even such a small difference will lead to significant differences in the prediction of FLDs and differences in the prediction of strain distribution for critical sheet formed products.

The specimens with specified pre-strains were stretched using a MAN 1000 kN hydrostatic universal testing machine. The generated strain in the specimen was controlled photogrammetrically. The 210 mm long tensile specimens were taken out of the pre-stretched regions of the sheets (Fig. 32) and underwent the same conditions as the uniaxial pre-strained specimens. Finally, the tensile test was performed on the samples to determine the mechanical properties of the pre-strained regions and thus ascertain the effect of pre-straining on strengthening.

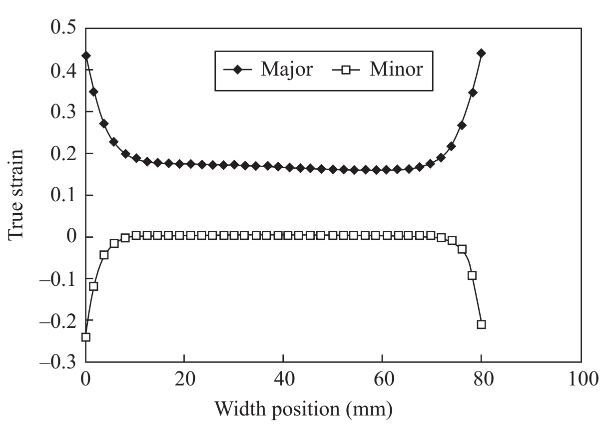

Fig. 31 Axial and transverse strain distribution in the plane strain test.

The change in length and the corresponding degree of pre-strain for the DP steel for each type of strain path are listed in Table 9.

Results and Discussion

Uniaxial condition

Fig. 33 shows the mechanical properties of DP steel conditioned with uniaxial pre-strains of up to 10% and aging temperatures of 100°C, 170°C, and 240°C with a holding time of 20 min. The strengthening effect is stronger with increasing aging temperatures which improve the diffusion conditions. The pre-straining influences the strengthening values such that the strength increase becomes significantly higher for a higher degree of pre-straining (Fig. 33A, B). It can be seen that for all aging temperatures, the yield and tensile strength increase exhibit a moderately linear dependence on the degree of pre-strain. Total elongation decreases with increasing pre-strain and aging temperature (Fig. 33C).

Biaxial condition

Fig. 34 shows the mechanical behavior of the DP-steel pre-strained under biaxial conditions with different degrees of pre-strain. Similar to the pre-strain conditions, the yield and tensile strength increases exhibit a moderately linear dependence on the degree of pre-strain at all aging temperature levels (Fig. 34A–B). In general, the strength level increases with increasing pre-strain and temperature. The highest strength but lowest ductility (Fig. 34C) was observed for the condition with the highest degree of pre-strain aged at 240°C. Comparing the uniaxial and biaxial conditions, it can be seen that a higher strength level with lower ductility is found for the biaxial condition.

Plane strain condition

Fig. 35 shows the mechanical properties for pre-strained specimens under plane strain conditions at different aging temperatures. The yield strength (Fig. 35A) and tensile strength (Fig. 35B) initially increase with pre-strains of up to 2% and reach a maximum value within the range of the investigation area. For pre-strains above 2%, tensile and yield strength values remain almost constant. In specimens pre-strained up to 2%, there is a significant increase in yield and tensile strengths beyond the first stage of aging at higher temperatures. This can be due to clustering of solute atoms or precipitation of low temperature carbides.[20] The most important aspect of these results is that the maximum increase in yield and tensile strengths at the completion of the first stage of aging is the same for all aging conditions. Total elongation decreases by increasing degree of pre-strain and aging temperature (Fig. 35C).

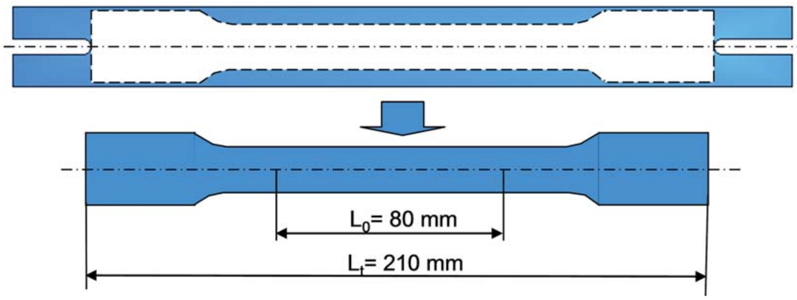

Fig. 32 Tensile specimen taken from pre-strained region in plane strain state.

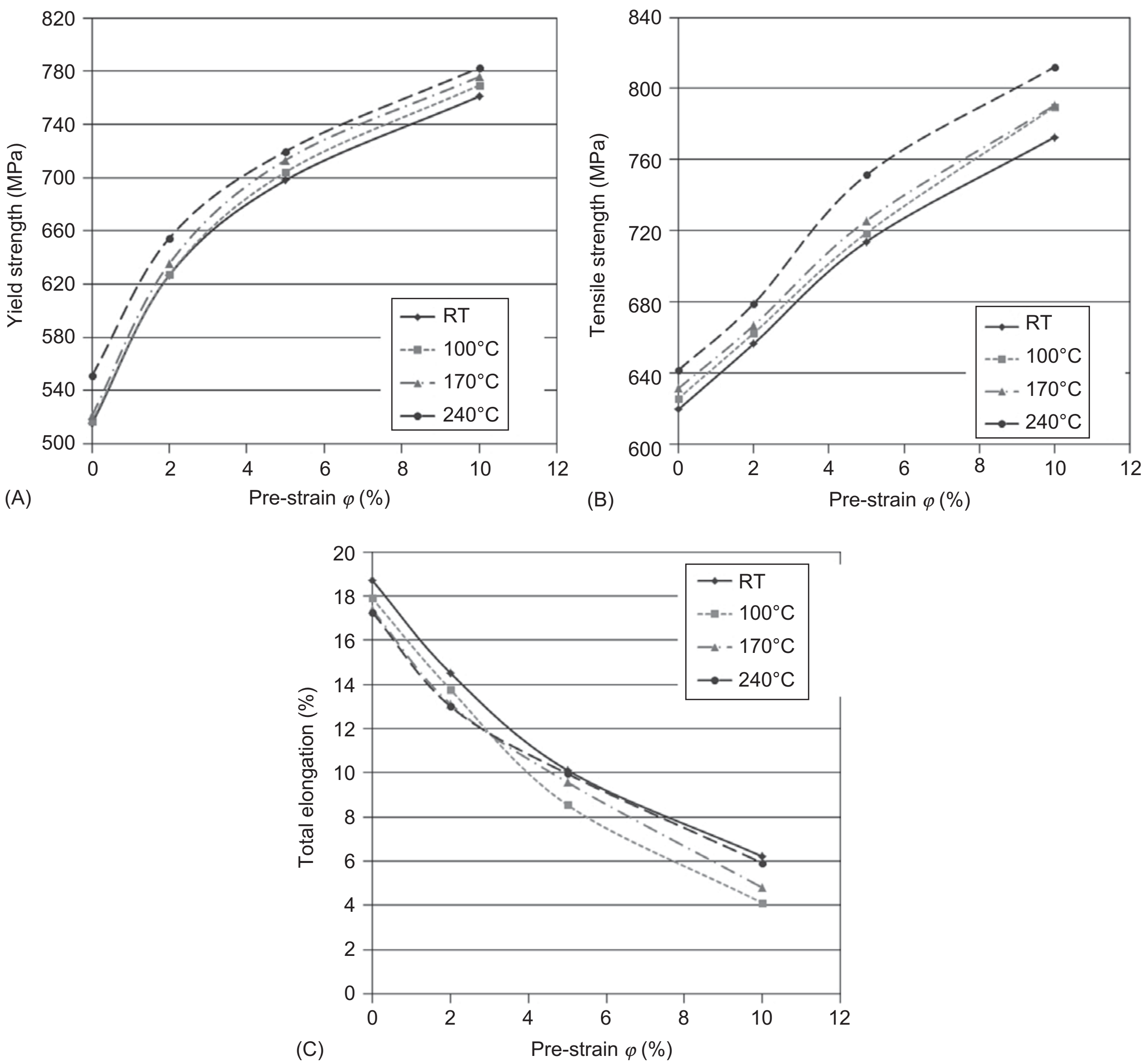

Fig. 33 Influence of pre-strain (uniaxial) and aging temperature on the mechanical properties of DP steel. Holding time: 20 min.

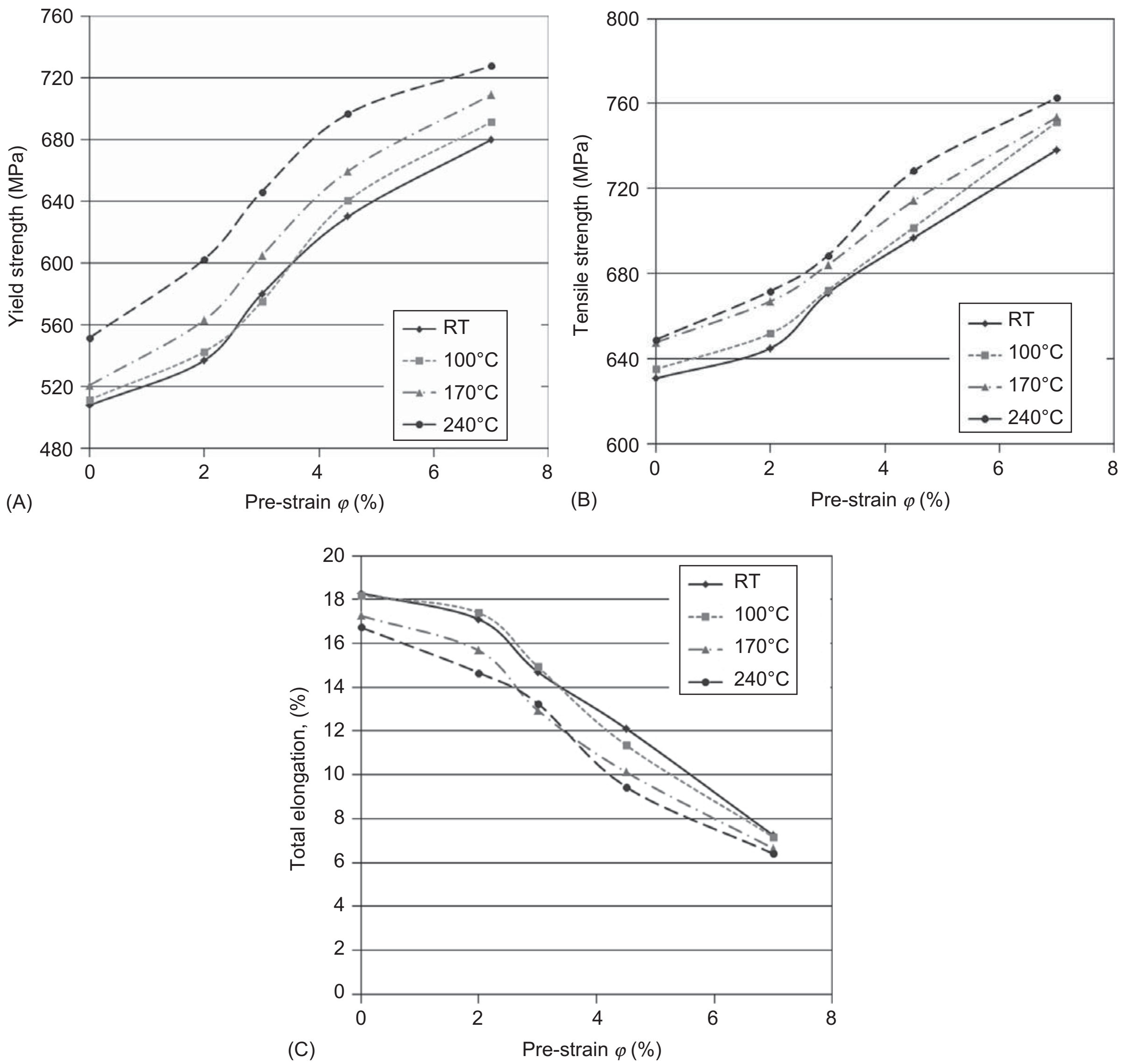

Fig. 34 Influence of pre-strain (biaxial) and aging temperature on the mechanical properties of DP steel. Holding time: 20 min.

The results reveal that up to saturation the amount of pre-strain does not influence the increase in yield stress during the formation of the atmosphere. Most importantly, at saturation, the degree of atmosphere formation is the same for all the pre-strain levels. Hence, the strengthening resulting from such atmosphere formation was found to be the same for all degrees of pre-strain.

For all pre-straining conditions, an aging treatment at 100°C exhibited no influence on the mechanical properties. Aging at temperatures of 170°C and 240°C led to an increase of the yield and tensile strengths and a decrease of the total elongation. This can be due to clustering of solute atoms or precipitation carbides at high temperature.

In this chapter, the authors presented a general overview of the influence of load path on BH response by showing mechanical properties for different load paths. The correlation between microstructure, mechanical properties, and different load paths was studied and published by the authors as well.[36,41]

Conclusion

1. The local mechanical strength behavior regarding the BH effect depend on the type of load path, degree of pre-strain, and bake hardening conditions.
2. Considering this effect and the differences of uniaxial, biaxial, and plane strain pre-strain, the type of the load path is of importance. A higher strength level can be

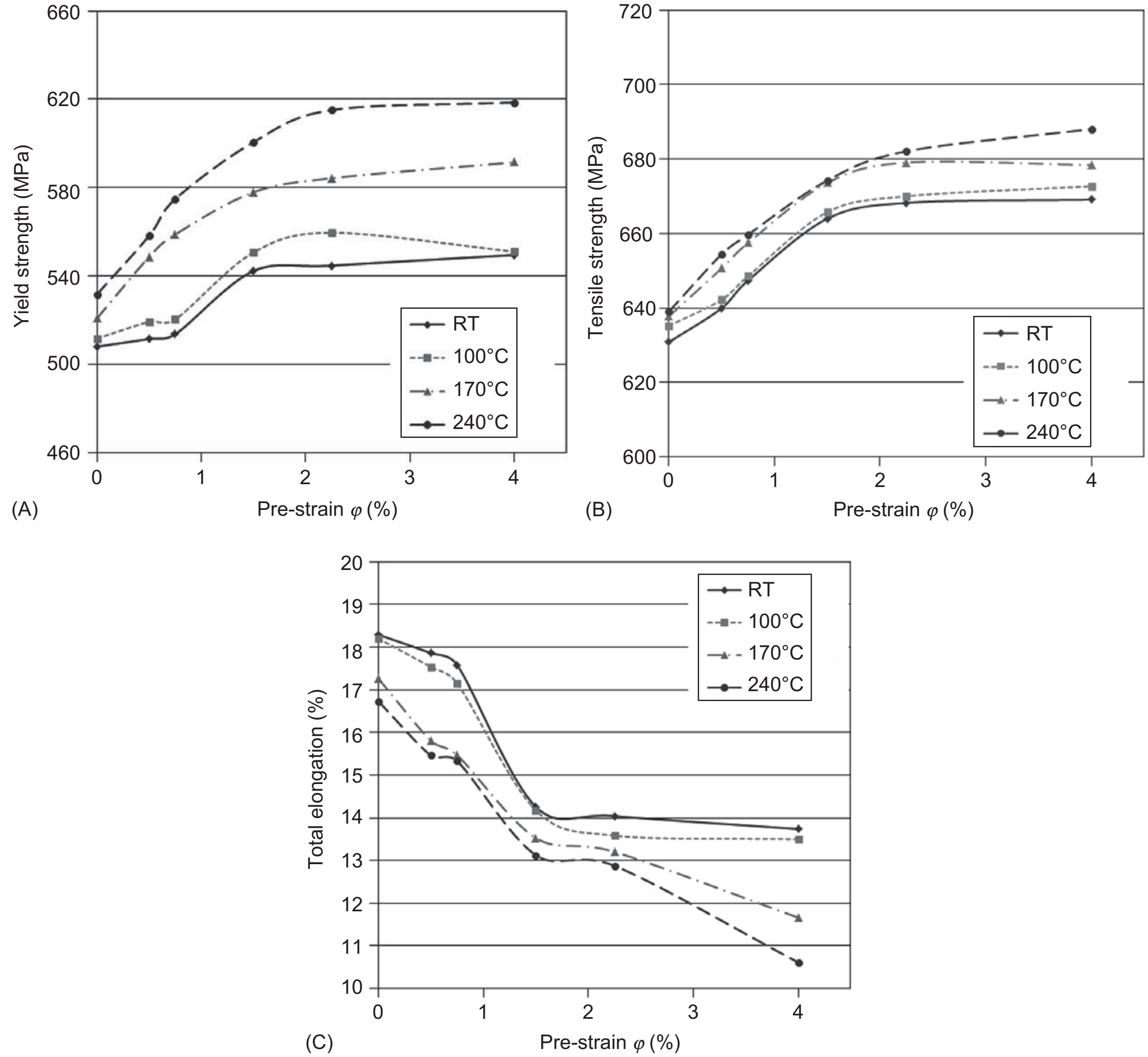

Fig. 35 Influence of plane strain pre-straining and aging temperature on mechanical properties of DP steel. Holding time: 20 min.

observed for biaxial condition at the same degree of pre-strain.

3. For all load conditions, a noticeable higher strength increment is observed after baking at 170°C and 240°C for 20 min. This increment is higher for the baking condition with 240°C and 20 min.

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